

ORIGINAL ARTICLE

KUMAP: Kernel Uniform Manifold Approximation and Projection for Medical Image Classification

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Received: 20 August 2025 | **Revised:** 23 April 2026 | **Accepted:** 26 April 2026

Keywords: feature extraction | kernel methods | medical image classification | out-of-sample embedding problem | UMAP

ABSTRACT

Uniform Manifold Approximation and Projection (UMAP) is a prominent dimensionality reduction and visualisation algorithm that has been widely adopted in the medical and biological fields. UMAP excels at capturing complex nonlinear relationships between high-dimensional and low-dimensional manifolds; however, it inherently suffers from the *Out-of-Sample embedding problem*, which prevents it from directly projecting new, unseen samples onto a previously learned manifold. In this paper, we propose *Kernel Uniform Manifold Approximation and Projection (KUMAP)*, a kernel-based extension designed to address this limitation. KUMAP maps training samples into a kernel space and learns the nonlinear mapping between this space and the low-dimensional embedding. Consequently, real-time dimensionality reduction for new samples is achieved through their projection into the kernel space. Furthermore, to leverage available label information, we introduce *Supervised Kernel Uniform Manifold Approximation and Projection (SKUMAP)*. Given that medical image analysis is critical for clinical diagnosis, dimensionality reduction techniques are essential for extracting key features to support intelligent decision-making. In this study, we evaluate the KUMAP and SKUMAP algorithms across eight medical image datasets to assess their effectiveness in resolving the Out-of-Sample embedding problem and extracting meaningful features. Experimental results demonstrate that KUMAP and SKUMAP successfully overcome the inherent limitations of UMAP while efficiently extracting essential features from complex medical imagery.

1 | Introduction

With the rapid advancement of machine learning and deep learning technologies, the accuracy of medical image classification has improved significantly. As a vital tool for clinical diagnosis, these automated systems enhance both the efficiency and precision of physicians' workflows. However, the prevalence of noise in medical images remains a critical challenge. To address this, effective dimensionality reduction methods for feature extraction are essential, as they can substantially boost the performance of classification models.

Linear dimensionality reduction algorithms operate on the assumption that the mapping from high-dimensional to low-dimensional space is linear. Principal component analysis (PCA) (Pearson 1901) is a quintessential example, alongside other notable techniques such as Neighbourhood Preserving Embedding (NPE) (He et al. 2005) and Locality Preserving Projections (LPP) (He and Niyogi 2004). Owing to their computational efficiency, strong interpretability and robust generalisation capabilities, these linear methods were initially favoured by researchers for feature extraction and images classification.

However, image data frequently exhibit intricate features, where linear mappings often prove inadequate for capturing the complex correlations between high-dimensional images and their low-dimensional projections. Consequently, nonlinear dimensionality reduction methods are essential for effective feature extraction from medical imagery. Prominent manifold learning techniques (Wang et al. 2025; Huang et al. 2024), such as Isometric Mapping (Isomap) (Tenenbaum et al. 2000), Locally Linear Embedding (LLE) (Roweis and Saul 2000), Laplacian Eigenmaps (LE) (Belkin and Niyogi 2003), Local Tangent Space Alignment (LTSA) (Zhang and Zha 2004), t-Distributed Stochastic Neighbour Embedding (t-SNE) (Maaten and Hinton 2008) and Uniform Manifold Approximation and Projection (UMAP) (McInnes et al. 2018; Leland et al. 2018), facilitate dimensionality reduction by preserving the intrinsic local geometric structure of the data. These methods have been extensively utilised in tasks including feature extraction, clustering support and high-dimensional data visualisation.

UMAP, a manifold learning method based on Riemannian geometry and algebraic topology, was introduced by McInnes et al. in 2018 (McInnes et al. 2018; Leland et al. 2018). It has been widely applied in fields such as bioinformatics (Becht et al. 2019; Yang et al. 2021; Karimzadeh et al. 2018), environmental science (Milošević et al. 2022; Kumari et al. 2021; Liu et al. 2023) and medicine (Sator et al. 2019; Du et al. 2023). Compared to earlier manifold learning methods like LLE (Roweis and Saul 2000), LE (Belkin and Niyogi 2003) and t-SNE (Maaten and Hinton 2008), UMAP offers lower computational complexity and improved feature extraction capabilities. Studies have demonstrated the advantages of UMAP in tasks such as feature extraction (Li et al. 2022; Taghizadeh-Mehrjardi et al. 2024; Weijler et al. 2021) and clustering support (Hozumi et al. 2021; Xia et al. 2021; Yang et al. 2022). Regrettably, UMAP suffers from the *Out-of-Sample (OOS) embedding problem* (Strange and Zwiggelaar 2011; Long and Ferguson 2019): *due to its reliance on implicit mapping, the algorithm cannot directly project novel, unseen points onto the pre-existing low-dimensional manifold*. This limitation significantly constrains its utility across broader applications. Consequently, there is an urgent need for a robust and efficient framework with an explicit mapping to address the OOS challenge within the UMAP architecture.

Currently, there are three primary theoretical frameworks for addressing the OOS embedding problem in manifold learning (Taşkın and Crawford 2019). The linearisation framework (He et al. 2005; He and Niyogi 2004; Saul and Roweis 2003) offers a straightforward approach; however, its overly simplistic assumptions often lead to suboptimal performance. Alternatively, the regression-based framework (Qiao et al. 2012; Liu et al. 2015; Sainburg et al. 2021) seeks to learn an explicit nonlinear mapping between high-dimensional and low-dimensional manifolds to achieve superior embedding results, though it suffers from significant computational overhead. Parametric UMAP (PUMAP) (Sainburg et al. 2021) was the first to address the OOS challenge by integrating the UMAP loss function as a regularisation term within a deep neural network, thereby learning an explicit mapping. Nevertheless, this approach entails significant computational overhead. In contrast, the kernel-based

framework (Bengio et al. 2003; Gisbrecht et al. 2015) combines the advantages of both preceding methods, capturing the nonlinear relationship between manifolds through a computationally efficient mapping formulation.

To efficiently address the OOS embedding problem in UMAP, we propose the Kernel Uniform Manifold Approximation and Projection (KUMAP) algorithm. As a representative kernel-based manifold learning method, KUMAP inherits the advantages of kernel methods in capturing nonlinear structures. However, unlike traditional kernel methods that reformulate the original optimisation problem in the kernel space, KUMAP captures complex local geometric features by learning a mapping matrix $\mathbf{A}$ between the kernel space and the low-dimensional manifold. Specifically, after projecting training samples into the kernel space, a nonlinear mapping is learned via the matrix $\mathbf{A}$. For OOS instances, KUMAP bypasses global manifold re-optimisation; instead, it projects these samples into the pre-constructed kernel space and applies matrix $\mathbf{A}$ to obtain their low-dimensional embeddings. Furthermore, to leverage label information, we extend this framework to a supervised variant, termed Supervised Kernel Uniform Manifold Approximation and Projection (SKUMAP). The overall framework of KUMAP and SKUMAP is illustrated in Figure 1.

The remainder of this paper is organised as follows: In Section 2, we review the related work; Section 3 provides a detailed exposition of the proposed KUMAP and SKUMAP algorithms; Section 4 reports the experimental results of KUMAP and SKUMAP on medical image datasets and Section 5 concludes the paper.

2 | Related Work

2.1 | UMAP

The training process of UMAP (McInnes et al. 2018; Leland et al. 2018) can be partitioned into two distinct phases: *Graph Construction* and *Graph Layout*. Figure 2 illustrates the fundamental concept of UMAP. In Figure 2a, UMAP constructs a k -nearest neighbour graph on the high-dimensional manifold to represent the topological structure of the original sample set. Subsequently, as shown in Figure 2b, UMAP optimises the layout of the embeddings on the low-dimensional manifold, aiming to faithfully preserve the topological relationships established in the high-dimensional space. This optimisation process ultimately achieves effective dimensionality reduction by aligning the low-dimensional structure with its high-dimensional counterpart.

2.1.1 | Graph Construction

For each sample point $\mathbf{x}_i$ in the high-dimensional manifold dataset $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$, its k -nearest neighbours, denoted as $\{\mathbf{x}_i^1, \mathbf{x}_i^2, \dots, \mathbf{x}_i^k\}$, are first identified. The similarity ω_{ij} between the sample point $\mathbf{x}_i$ and its j th neighbour $\mathbf{x}_i^j$ is defined as follows:

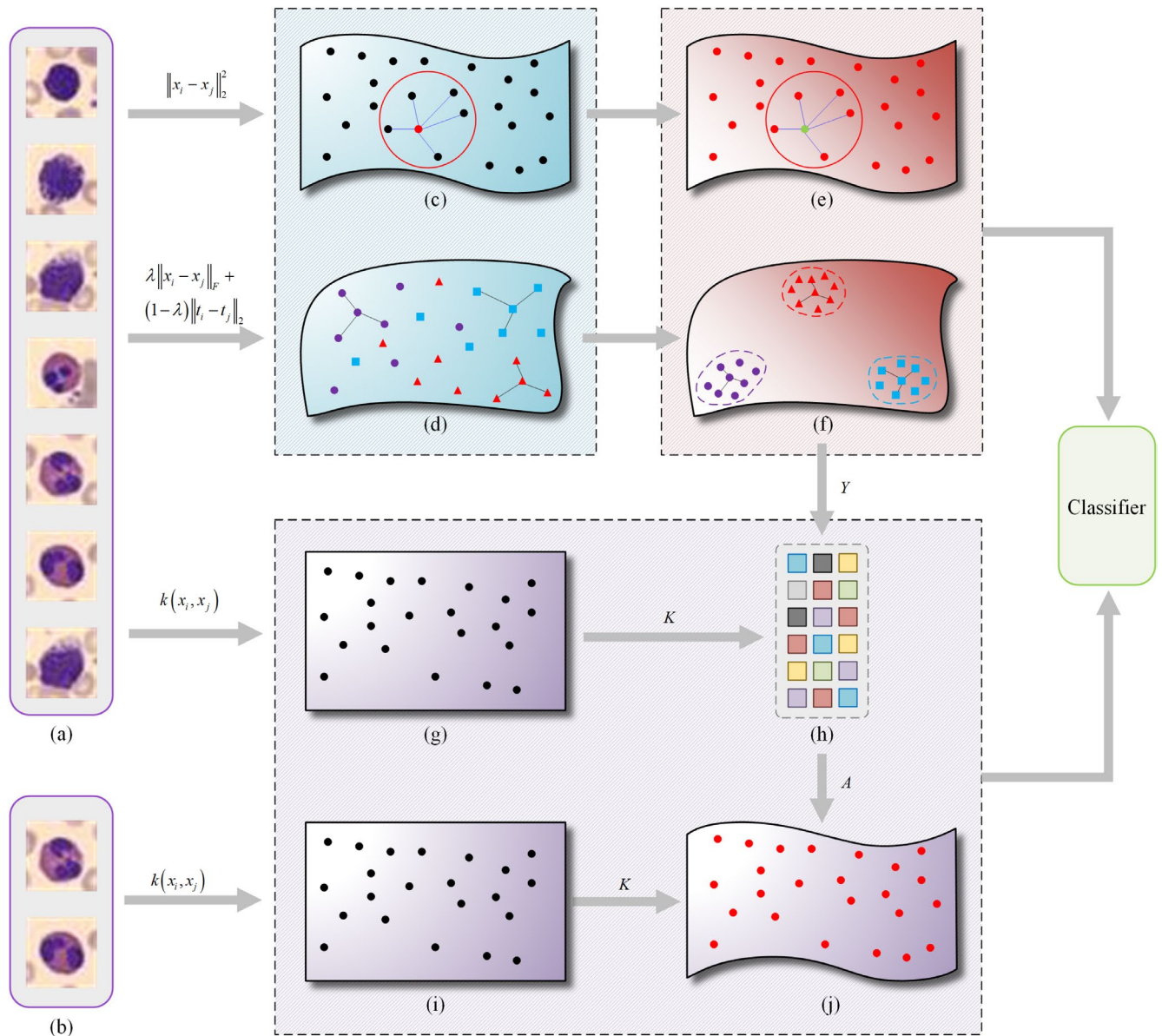


FIGURE 1 | Fundamental Concepts of KUMAP and SKUMAP: (a) Training samples; (b) out-of-sample data; (c) the process of graph construction on the high-dimensional manifold in KUMAP; (d) the process of graph construction on the high-dimensional manifold in SKUMAP; (e) the optimisation process of projection points on the low-dimensional manifold in KUMAP; (f) the optimisation process of projection points on the low-dimensional manifold in SKUMAP; (g) training samples in the kernel space; (h) the mapping relationship between the kernel space and the low-dimensional manifold; (i) out-of-sample data in the kernel space; (j) out-of-sample samples on the low-dimensional manifold.

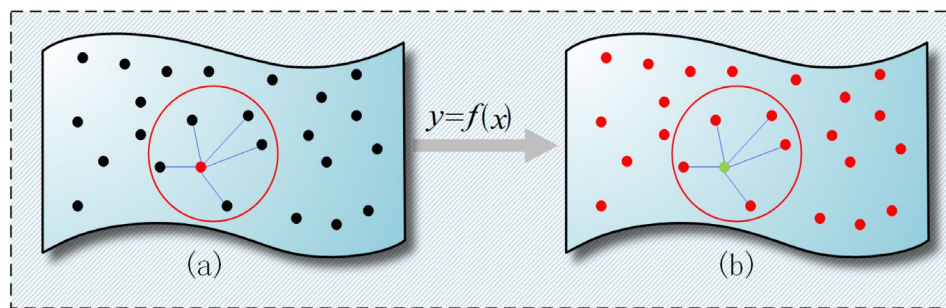


FIGURE 2 | Fundamental concepts of UMAP: (a) The process of graph construction on the high-dimensional manifold; (b) the optimisation process of projection points on the low-dimensional manifold.

$$\omega(\mathbf{x}_i, \mathbf{x}_i^j) = \exp\left(\frac{-\max(0, d(\mathbf{x}_i, \mathbf{x}_i^j) - \rho_i)}{\sigma_i}\right) \quad (1)$$

where ρ_i denotes the distance from the sample point $\mathbf{x}_i$ to its nearest neighbour. Specifically, ρ_i is defined as follows:

$$\rho_i = \min \left\{ d(\mathbf{x}_i, \mathbf{x}_i^j) \mid 1 \leq j \leq k, d(\mathbf{x}_i, \mathbf{x}_i^j) > 0 \right\} \quad (2)$$

σ_i denotes the local scale parameter for the sample point $\mathbf{x}_i$, which is typically determined via a binary search to satisfy specific local connectivity constraints. Its implicit relationship can be expressed as:

$$\sum_{j=1}^k \exp\left(\frac{-\max(0, d(\mathbf{x}_i, \mathbf{x}_i^j) - \rho_i)}{\sigma_i}\right) = \log_2(k) \quad (3)$$

2.1.2 | Graph Layout

In the low-dimensional UMAP embedding, the edge weight v_{ij} between points $\mathbf{y}_i$ and $\mathbf{y}_j$ (corresponding to the high-dimensional samples $\mathbf{x}_i$ and $\mathbf{x}_j$, respectively) is defined as follows:

$$v(\mathbf{y}_i, \mathbf{y}_j) = \left(1 + a(\|\mathbf{y}_i - \mathbf{y}_j\|_2^2)^b\right)^{-1} \quad (4)$$

UMAP achieves dimensionality reduction by minimising the fuzzy set cross-entropy function, thereby reducing the discrepancy between the high-dimensional similarities $\omega(\mathbf{x}_i, \mathbf{x}_j)$ and their low-dimensional counterparts $v(\mathbf{y}_i, \mathbf{y}_j)$. This optimisation process continuously aligns the topological structure of the low-dimensional embeddings with that of the original sample set in the high-dimensional manifold.

$$\text{CE} = \sum \omega \cdot \log\left(\frac{\omega}{v}\right) + (1 - \omega) \cdot \log\left(\frac{1 - \omega}{1 - v}\right) \quad (5)$$

TABLE 1 | Key technical parameters of the datasets.

Datasets	Samples	Classes	Channels	Dimensions
BloodMNIST	17,092	8	3	2352
DermaMNIST	10,015	7	3	2352
OCTMNIST	109,309	4	1	784
OrganAMNIST	58,830	11	1	784
OrganCMNIST	23,583	11	1	784
OrganSMNIST	25,211	11	1	784
PneumoniaMNIST	5856	2	1	784
RetinaMNIST	1600	7	3	2352

TABLE 2 | Properties of KUMAP, SKUMAP and baselines.

Methods	Type	OOS	Supervision	Mapping	Specificity parameters
LDA	Linear DR	✓	Discriminant	Linear	$n_components = 2$
KLDA	Kernel DR	✓	Discriminant	NonLinear	$n_components = 2, kernel = rbf$
NCA	Metric Learning	✓	Supervised	Linear	$n_components = 2$
SITML	Metric Learning	✓	Supervised	Linear	$n_constraints = 200$
PaCMAP	Manifold Learning	✗	Unsupervised	NonLinear	None
TriMAP	Manifold Learning	✗	Unsupervised	NonLinear	None
UMATO	Manifold Learning	✗	Unsupervised	NonLinear	None
MDE	Manifold Learning	✗	Unsupervised	NonLinear	$device = GPU$
UMAP	Manifold Learning	✗	Unsupervised	NonLinear	None
PUMAP	Manifold Learning	✓	Unsupervised	NonLinear	None
KUMAP	Manifold Learning	✓	Unsupervised	NonLinear	None
SKUMAP	Manifold Learning	✓	Supervised	NonLinear	None

Specifically, in the low-dimensional manifold, UMAP optimises the projected point set by iteratively applying attractive force $\mathcal{F}_{UMAP} = \nabla \omega \log v$ and repulsive force $\mathcal{H}_{UMAP} = \nabla (1 - \omega) \log(1 - v)$ between projected points $\mathbf{y}_i$ and $\mathbf{y}_j$.

2.2 | Kernel-Based Dimensionality Reduction

Kernel-based dimensionality reduction methods comprise a suite of techniques, including Kernel Principal Component Analysis (KPCA) (Scholkopf et al. 1998), Kernel Canonical Correlation Analysis (KCCA) (Hardoon et al. 2004) and Kernel Linear Discriminant Analysis (KLDA) (Mika et al. 1999; Baudat and Anouar 2000). These algorithms leverage kernel functions to project sample points into a high-dimensional feature space (kernel space), where dimensionality reduction is subsequently performed. By operating within this kernel space, these methods

can capture intricate nonlinear structures that are otherwise inaccessible to their linear counterparts, thereby significantly enhancing feature extraction capabilities. Standard choices for these transformations include linear, polynomial, Gaussian and Laplacian kernel functions.

3 | Methods

To address the Out-of-Sample embedding problem in UMAP and capture the nonlinear mapping between high-dimensional and low-dimensional manifolds, this section introduces the KUMAP algorithm, developed within the kernelisation framework of manifold learning. KUMAP aims to map high-dimensional samples into a Reproducing Kernel Hilbert Space (RKHS) and perform dimensionality reduction by optimising a transformation matrix $\mathbf{A}$ that links the kernel space to the low-dimensional manifold.

TABLE 3 | Average classification accuracy ($\pm$ standard deviation) of 12 algorithms across 8 datasets.

Methods	BloodMNIST	DermaMNIST	OCTMNIST	OrganAMNIST
LDA	0.5015 $\pm$ 0.0062	0.3667 $\pm$ 0.0256	0.5481 $\pm$ 0.0026	0.4659 $\pm$ 0.0039
KLDA	0.6343 $\pm$ 0.0078	0.5203 $\pm$ 0.0161	0.7301 $\pm$ 0.0012	0.4219 $\pm$ 0.0309
PaCMAP	0.5649 $\pm$ 0.0046	0.5687 $\pm$ 0.0065	0.4152 $\pm$ 0.0029	0.8974 $\pm$ 0.0021
TriMAP	0.5696 $\pm$ 0.0032	0.5687 $\pm$ 0.0050	0.3879 $\pm$ 0.0027	0.8290 $\pm$ 0.0022
UMATO	0.5269 $\pm$ 0.0030	0.5483 $\pm$ 0.0056	0.3996 $\pm$ 0.0019	0.5954 $\pm$ 0.0029
MDE	0.5490 $\pm$ 0.0041	0.5685 $\pm$ 0.0047	0.4035 $\pm$ 0.0024	0.8247 $\pm$ 0.0029
UMAP	0.5809 $\pm$ 0.0070	0.5777 $\pm$ 0.0042	0.4167 $\pm$ 0.0023	0.8627 $\pm$ 0.0042
NCA	0.5718 $\pm$ 0.0114	0.3573 $\pm$ 0.0348	—	0.5357 $\pm$ 0.0020
SITML	0.5262 $\pm$ 0.0181	0.6532 $\pm$ 0.0053	<u>0.6466 $\pm$ 0.0047</u>	0.7897 $\pm$ 0.0066
PUMAP	0.4988 $\pm$ 0.0071	0.5538 $\pm$ 0.0087	0.3957 $\pm$ 0.0015	0.6011 $\pm$ 0.0101
KUMAP	<u>0.6130 $\pm$ 0.0050</u>	0.6245 $\pm$ 0.0025	0.5433 $\pm$ 0.0020	<u>0.9537 $\pm$ 0.0013</u>
SKUMAP	0.6423 $\pm$ 0.0177	<u>0.6333 $\pm$ 0.0074</u>	0.5431 $\pm$ 0.0013	0.9554 $\pm$ 0.0039
Methods	OrganCMNIST	OrganSMNIST	PneumoniaMNIST	RetinaMNIST
LDA	0.4067 $\pm$ 0.0177	0.2967 $\pm$ 0.0031	0.8399 $\pm$ 0.0110	0.4139 $\pm$ 0.0177
KLDA	0.2665 $\pm$ 0.0159	0.2106 $\pm$ 0.0136	0.9267 $\pm$ 0.0046	0.3418 $\pm$ 0.0305
PaCMAP	0.8118 $\pm$ 0.0024	0.6481 $\pm$ 0.0028	0.8754 $\pm$ 0.0040	0.4034 $\pm$ 0.0121
TriMAP	0.7886 $\pm$ 0.0031	0.5973 $\pm$ 0.0042	0.8804 $\pm$ 0.0051	0.3884 $\pm$ 0.0095
UMATO	0.5291 $\pm$ 0.0030	0.4087 $\pm$ 0.0131	0.8844 $\pm$ 0.0070	0.3758 $\pm$ 0.0155
MDE	0.6571 $\pm$ 0.0025	0.4999 $\pm$ 0.0053	0.8900 $\pm$ 0.0059	0.4055 $\pm$ 0.0181
UMAP	0.7007 $\pm$ 0.0040	0.5443 $\pm$ 0.0026	0.8897 $\pm$ 0.0060	0.4012 $\pm$ 0.0077
NCA	0.4929 $\pm$ 0.0032	0.3452 $\pm$ 0.0034	0.9217 $\pm$ 0.0046	0.4233 $\pm$ 0.0138
SITML	0.4063 $\pm$ 0.0079	0.3733 $\pm$ 0.0058	<u>0.9222 $\pm$ 0.0058</u>	0.4573 $\pm$ 0.0153
PUMAP	0.4652 $\pm$ 0.0134	0.3466 $\pm$ 0.0098	0.8622 $\pm$ 0.0108	0.3614 $\pm$ 0.0140
KUMAP	<u>0.8513 $\pm$ 0.0011</u>	<u>0.7601 $\pm$ 0.0022</u>	0.8994 $\pm$ 0.0025	0.4228 $\pm$ 0.0126
SKUMAP	0.8655 $\pm$ 0.0105	0.7669 $\pm$ 0.0122	0.9041 $\pm$ 0.0041	<u>0.4401 $\pm$ 0.0125</u>

Note: In each column, the best performance is highlighted in bold and the second-best is underlined.

TABLE 4 | Average classification precision ($\pm$ standard deviation) of 12 algorithms 8 datasets.

Methods	BloodMNIST	DermaMNIST	OCTMNIST	OrganAMNIST
LDA	0.4467 $\pm$ 0.0072	0.1687 $\pm$ 0.0042	0.4227 $\pm$ 0.0028	0.4117 $\pm$ 0.0026
KLDA	<u>0.6001 $\pm$ 0.0100</u>	0.2498 $\pm$ 0.0155	0.6048 $\pm$ 0.0016	0.4139 $\pm$ 0.0234
PaCMAP	0.5381 $\pm$ 0.0053	0.2193 $\pm$ 0.0058	0.3000 $\pm$ 0.0022	0.8900 $\pm$ 0.0026
TriMAP	0.5438 $\pm$ 0.0032	0.2135 $\pm$ 0.0043	0.2772 $\pm$ 0.0013	0.8263 $\pm$ 0.0025
UMATO	0.5106 $\pm$ 0.0037	0.2014 $\pm$ 0.0071	0.2910 $\pm$ 0.0013	0.6189 $\pm$ 0.0059
MDE	0.5289 $\pm$ 0.0047	0.2145 $\pm$ 0.0063	0.2923 $\pm$ 0.0017	0.8203 $\pm$ 0.0035
UMAP	0.5609 $\pm$ 0.0081	0.2263 $\pm$ 0.0058	0.3055 $\pm$ 0.0016	0.8583 $\pm$ 0.0042
NCA	0.5231 $\pm$ 0.0098	0.1745 $\pm$ 0.0061	—	0.5038 $\pm$ 0.0043
SITML	0.5523 $\pm$ 0.0149	0.3271 $\pm$ 0.0082	<u>0.5496 $\pm$ 0.0053</u>	0.8151 $\pm$ 0.0119
PUMAP	0.4799 $\pm$ 0.0073	0.2017 $\pm$ 0.0061	0.2874 $\pm$ 0.0016	0.5981 $\pm$ 0.0079
KUMAP	0.5898 $\pm$ 0.0061	0.2828 $\pm$ 0.0043	0.4273 $\pm$ 0.0024	<u>0.9497 $\pm$ 0.0012</u>
SKUMAP	0.6206 $\pm$ 0.0185	<u>0.2923 $\pm$ 0.0059</u>	0.4269 $\pm$ 0.0012	0.9518 $\pm$ 0.0040
Methods	OrganCMNIST	OrganSMNIST	PneumoniaMNIST	RetinaMNIST
LDA	0.3601 $\pm$ 0.0149	0.2731 $\pm$ 0.0032	0.7971 $\pm$ 0.0148	0.2938 $\pm$ 0.0141
KLDA	0.3195 $\pm$ 0.0339	0.2695 $\pm$ 0.0075	<u>0.9101 $\pm$ 0.0109</u>	0.2719 $\pm$ 0.0233
PaCMAP	0.7888 $\pm$ 0.0032	0.6137 $\pm$ 0.0042	0.8426 $\pm$ 0.0056	0.2835 $\pm$ 0.0124
TriMAP	0.7678 $\pm$ 0.0034	0.5678 $\pm$ 0.0050	0.8481 $\pm$ 0.0074	0.2691 $\pm$ 0.0097
UMATO	0.5668 $\pm$ 0.0046	0.3976 $\pm$ 0.0122	0.8538 $\pm$ 0.0087	0.2720 $\pm$ 0.0160
MDE	0.6641 $\pm$ 0.0040	0.4846 $\pm$ 0.0050	0.8597 $\pm$ 0.0088	0.2817 $\pm$ 0.0133
UMAP	0.7007 $\pm$ 0.0049	0.5236 $\pm$ 0.0021	0.8597 $\pm$ 0.0082	0.2790 $\pm$ 0.0106
NCA	0.4642 $\pm$ 0.0049	0.3277 $\pm$ 0.0067	0.8992 $\pm$ 0.0061	0.2953 $\pm$ 0.0147
SITML	0.6399 $\pm$ 0.0136	0.5959 $\pm$ 0.0152	0.9165 $\pm$ 0.0080	0.3215 $\pm$ 0.0059
PUMAP	0.4821 $\pm$ 0.0117	0.3358 $\pm$ 0.0107	0.8271 $\pm$ 0.0160	0.2538 $\pm$ 0.0092
KUMAP	<u>0.8365 $\pm$ 0.0016</u>	<u>0.7281 $\pm$ 0.0021</u>	0.8742 $\pm$ 0.0043	0.2935 $\pm$ 0.0169
SKUMAP	0.8512 $\pm$ 0.0103	0.7351 $\pm$ 0.0126	0.8800 $\pm$ 0.0080	<u>0.3025 $\pm$ 0.0124</u>

Note: In each column, the best performance is highlighted in bold and the second-best is underlined.

3.1 | Kernel UMAP

For the training dataset $\mathbf{X}'$, the low-dimensional projection $\mathbf{y}(\mathbf{x}_i)$ of each point $\mathbf{x}_i$ residing on the high-dimensional manifold is first obtained via the conventional UMAP algorithm. Simultaneously, these training samples are mapped into a kernel space using a kernel function $k(\mathbf{x}_i, \mathbf{x}_j)$, followed by the construction of a normalised kernel matrix $\mathbf{K}$ as follows:

$$\mathbf{K}_{ij} = \frac{k(\mathbf{x}_i, \mathbf{x}_j)}{\sum_m k(\mathbf{x}_i, \mathbf{x}_m)} \quad (6)$$

The coefficient vector α_i characterises the nonlinear mapping between the kernel space representations and their low-dimensional counterparts. Based on the generalised kernelisation framework for manifold learning defined in Equation (7),

the projection $\mathbf{y}(\mathbf{x}_0)$ for an unseen out-of-sample point $\mathbf{x}_0$ can be efficiently computed using the optimised coefficient matrix $\mathbf{A}$:

$$\mathbf{x}_0 \rightarrow \mathbf{y}(\mathbf{x}_0) = \sum_j \alpha_j \cdot \frac{k(\mathbf{x}_0, \mathbf{x}_j)}{\sum_m k(\mathbf{x}_0, \mathbf{x}_m)} \quad (7)$$

With the correspondence between the normalised kernel matrix $\mathbf{K}$ and the UMAP embeddings $\mathbf{Y}$ established, the objective shifts to determining the coefficient matrix $\mathbf{A}$. This is achieved by minimising the discrepancy between the kernel-derived projections and the standard UMAP embeddings, formulated as the following objective function:

$$\min_{\mathbf{A}} \mathcal{L}(\mathbf{A}) = \min_{\mathbf{A}} \sum_i \left\| \mathbf{y}_i - \mathbf{y}(\mathbf{x}_i) \right\|^2 = \min_{\mathbf{A}} \left\| \mathbf{K}\mathbf{A} - \mathbf{Y} \right\|_F^2 \quad (8)$$

TABLE 5 | Average classification recall ($\pm$ standard deviation) of 12 algorithms across 8 datasets.

Methods	BloodMNIST	DermaMNIST	OCTMNIST	OrganAMNIST
LDA	0.4692 $\pm$ 0.0070	0.1857 $\pm$ 0.0096	0.4220 $\pm$ 0.0022	0.4119 $\pm$ 0.0026
KLDA	<u>0.5964 $\pm$ 0.0085</u>	0.2680 $\pm$ 0.0180	0.6094 $\pm$ 0.0021	0.4262 $\pm$ 0.0211
PaCMAP	0.5373 $\pm$ 0.0054	0.2200 $\pm$ 0.0067	0.2999 $\pm$ 0.0022	0.8906 $\pm$ 0.0022
TriMAP	0.5436 $\pm$ 0.0038	0.2131 $\pm$ 0.0047	0.2772 $\pm$ 0.0013	0.8270 $\pm$ 0.0023
UMATO	0.5123 $\pm$ 0.0054	0.2023 $\pm$ 0.0079	0.2910 $\pm$ 0.0013	0.6202 $\pm$ 0.0060
MDE	0.5279 $\pm$ 0.0045	0.2138 $\pm$ 0.0066	0.2922 $\pm$ 0.0016	0.8215 $\pm$ 0.0033
UMAP	0.5591 $\pm$ 0.0068	0.2271 $\pm$ 0.0051	0.3054 $\pm$ 0.0014	0.8594 $\pm$ 0.0046
NCA	0.5347 $\pm$ 0.0100	0.1935 $\pm$ 0.0152	—	0.5019 $\pm$ 0.0024
SITML	0.5182 $\pm$ 0.0171	0.2821 $\pm$ 0.0157	<u>0.5012 $\pm$ 0.0043</u>	0.7963 $\pm$ 0.0099
PUMAP	0.4765 $\pm$ 0.0089	0.2019 $\pm$ 0.0061	0.2875 $\pm$ 0.0016	0.5978 $\pm$ 0.0089
KUMAP	0.5936 $\pm$ 0.0042	0.2741 $\pm$ 0.0069	0.4213 $\pm$ 0.0021	<u>0.9520 $\pm$ 0.0019</u>
SKUMAP	0.6256 $\pm$ 0.0184	<u>0.2794 $\pm$ 0.0061</u>	0.4209 $\pm$ 0.0013	0.9542 $\pm$ 0.0040
Methods	OrganCMNIST	OrganSMNIST	PneumoniaMNIST	RetinaMNIST
LDA	0.3616 $\pm$ 0.0143	0.2682 $\pm$ 0.0039	0.7997 $\pm$ 0.0097	0.2918 $\pm$ 0.0121
KLDA	0.3172 $\pm$ 0.0207	0.2658 $\pm$ 0.0074	0.9054 $\pm$ 0.0102	0.2628 $\pm$ 0.0220
PaCMAP	0.7878 $\pm$ 0.0030	0.6133 $\pm$ 0.0035	0.8422 $\pm$ 0.0073	0.2838 $\pm$ 0.0111
TriMAP	0.7673 $\pm$ 0.0029	0.5669 $\pm$ 0.0039	0.8491 $\pm$ 0.0072	0.2692 $\pm$ 0.0097
UMATO	0.5650 $\pm$ 0.0027	0.3964 $\pm$ 0.0112	0.8531 $\pm$ 0.0110	0.2705 $\pm$ 0.0168
MDE	0.6631 $\pm$ 0.0015	0.4835 $\pm$ 0.0045	0.8620 $\pm$ 0.0065	0.2813 $\pm$ 0.0148
UMAP	0.6987 $\pm$ 0.0031	0.5235 $\pm$ 0.0027	0.8619 $\pm$ 0.0075	0.2779 $\pm$ 0.0099
NCA	0.4657 $\pm$ 0.0046	0.3279 $\pm$ 0.0051	<u>0.9034 $\pm$ 0.0066</u>	0.2924 $\pm$ 0.0140
SITML	0.4372 $\pm$ 0.0109	0.3915 $\pm$ 0.0072	0.8818 $\pm$ 0.0098	0.3169 $\pm$ 0.0075
PUMAP	0.4804 $\pm$ 0.0142	0.3374 $\pm$ 0.0089	0.8230 $\pm$ 0.0102	0.2517 $\pm$ 0.0096
KUMAP	<u>0.8383 $\pm$ 0.0017</u>	<u>0.7280 $\pm$ 0.0031</u>	0.8706 $\pm$ 0.0054	0.2928 $\pm$ 0.0167
SKUMAP	0.8521 $\pm$ 0.0109	0.7347 $\pm$ 0.0112	0.8765 $\pm$ 0.0082	<u>0.3022 $\pm$ 0.0102</u>

Note: In each column, the best performance is highlighted in bold and the second-best is underlined.

This formulation constitutes a convex quadratic optimisation problem. If $\mathbf{K}$ is non-singular, a closed-form solution is readily available via the ordinary least squares method:

$$\mathbf{A} = \mathbf{K}^{\dagger} \mathbf{Y} \quad (9)$$

However, in practical applications, particularly with medical image datasets, $\mathbf{K}$ is frequently singular or ill-conditioned. To ensure numerical stability and computational efficiency, the following strategies are implemented:

- *Moore-Penrose Pseudoinverse*: To resolve the potential non-invertibility of $\mathbf{K}$, the pseudoinverse is utilised, ensuring a stable and unique solution even in rank-deficient cases.
- *Block-wise Computation Strategy*: To mitigate the high memory overhead associated with large-scale kernel

matrices, a block-wise solver is adopted, thereby enhancing the algorithm's scalability for high-volume datasets.

3.2 | Kernel Function of KUMAP

Equation (10) defines the general form of the Laplacian kernel function, which incorporates a normalisation factor σ_i . In KUMAP, this kernel is utilised to project samples from the high-dimensional manifold into a kernel space, facilitating the subsequent training of the coefficient matrix $\mathbf{A}$.

$$k(\mathbf{x}, \mathbf{x}_i) = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}_i\|}{\sigma_i}\right) \quad (10)$$

TABLE 6 | Average classification $F1$ score ($\pm$ standard deviation) of 12 algorithms across 8 datasets.

Methods	BloodMNIST	DermaMNIST	OCTMNIST	OrganAMNIST
LDA	0.4480 $\pm$ 0.0065	0.1568 $\pm$ 0.0049	0.4223 $\pm$ 0.0024	0.4117 $\pm$ 0.0025
KLDA	<u>0.5963 $\pm$ 0.0090</u>	0.2540 $\pm$ 0.0158	0.6067 $\pm$ 0.0017	0.3891 $\pm$ 0.0262
PaCMAP	0.5375 $\pm$ 0.0053	0.2192 $\pm$ 0.0061	0.2999 $\pm$ 0.0022	0.8903 $\pm$ 0.0022
TriMAP	0.5435 $\pm$ 0.0032	0.2130 $\pm$ 0.0044	0.2772 $\pm$ 0.0013	0.8266 $\pm$ 0.0023
UMATO	0.5111 $\pm$ 0.0043	0.2015 $\pm$ 0.0074	0.2910 $\pm$ 0.0013	0.6194 $\pm$ 0.0058
MDE	0.5281 $\pm$ 0.0044	0.2139 $\pm$ 0.0065	0.2922 $\pm$ 0.0016	0.8208 $\pm$ 0.0032
UMAP	0.5598 $\pm$ 0.0072	0.2263 $\pm$ 0.0054	0.3054 $\pm$ 0.0015	0.8587 $\pm$ 0.0043
NCA	0.5262 $\pm$ 0.0100	0.1617 $\pm$ 0.0104	—	0.5027 $\pm$ 0.0029
SITML	0.4929 $\pm$ 0.0214	0.2872 $\pm$ 0.0130	<u>0.5138 $\pm$ 0.0045</u>	0.7860 $\pm$ 0.0106
PUMAP	0.4778 $\pm$ 0.0080	0.2015 $\pm$ 0.0061	0.2875 $\pm$ 0.0016	0.5977 $\pm$ 0.0082
KUMAP	0.5910 $\pm$ 0.0052	0.2768 $\pm$ 0.0057	0.4233 $\pm$ 0.0022	<u>0.9508 $\pm$ 0.0015</u>
SKUMAP	0.6201 $\pm$ 0.0179	<u>0.2830 $\pm$ 0.0044</u>	0.4229 $\pm$ 0.0012	0.9529 $\pm$ 0.0040
Methods	OrganCMNIST	OrganSMNIST	PneumoniaMNIST	RetinaMNIST
LDA	0.3606 $\pm$ 0.0146	0.2701 $\pm$ 0.0036	0.7978 $\pm$ 0.0109	0.2901 $\pm$ 0.0133
KLDA	0.2793 $\pm$ 0.0267	0.2366 $\pm$ 0.0120	0.9070 $\pm$ 0.0049	0.2563 $\pm$ 0.0213
PaCMAP	0.7881 $\pm$ 0.0028	0.6133 $\pm$ 0.0037	0.8423 $\pm$ 0.0054	0.2817 $\pm$ 0.0109
TriMAP	0.7673 $\pm$ 0.0031	0.5670 $\pm$ 0.0044	0.8485 $\pm$ 0.0066	0.2680 $\pm$ 0.0097
UMATO	0.5657 $\pm$ 0.0032	0.3967 $\pm$ 0.0117	0.8533 $\pm$ 0.0090	0.2696 $\pm$ 0.0159
MDE	0.6634 $\pm$ 0.0025	0.4837 $\pm$ 0.0045	0.8607 $\pm$ 0.0068	0.2798 $\pm$ 0.0139
UMAP	0.6995 $\pm$ 0.0038	0.5232 $\pm$ 0.0020	0.8608 $\pm$ 0.0076	0.2773 $\pm$ 0.0100
NCA	0.4646 $\pm$ 0.0045	0.3274 $\pm$ 0.0058	<u>0.9012 $\pm$ 0.0058</u>	0.2917 $\pm$ 0.0142
SITML	0.4361 $\pm$ 0.0076	0.4022 $\pm$ 0.0073	0.8969 $\pm$ 0.0079	0.3155 $\pm$ 0.0067
PUMAP	0.4806 $\pm$ 0.0127	0.3360 $\pm$ 0.0096	0.8247 $\pm$ 0.0121	0.2510 $\pm$ 0.0092
KUMAP	<u>0.8362 $\pm$ 0.0014</u>	<u>0.7264 $\pm$ 0.0027</u>	0.8723 $\pm$ 0.0029	0.2916 $\pm$ 0.0169
SKUMAP	0.8498 $\pm$ 0.0100	0.7329 $\pm$ 0.0112	0.8779 $\pm$ 0.0047	<u>0.3004 $\pm$ 0.0111</u>

Note: In each column, the best performance is highlighted in bold and the second-best is underlined.

For each sample $\mathbf{x}_i$ within the training set $\mathbf{X}'$, the normalisation factor σ_i is determined by the distance to its nearest neighbour. Specifically, σ_i is formulated as follows:

$$\sigma_i = 2 \cdot \beta^2 \cdot \min_{\mathbf{x} \in \mathbf{X}'} \|\mathbf{x} - \mathbf{x}_i\| \quad (11)$$

where β denotes a scaling factor. In both Kernel UMAP and Supervised Kernel UMAP algorithms, β is typically set to a small positive constant to ensure numerical stability and optimal performance. Based on empirical evaluation, we have assigned $\beta = 0.08$ for our experiments.

3.3 | Supervised Kernel UMAP

Both the traditional UMAP and the KUMAP method introduced in Section 3.1 are unsupervised dimensionality reduction techniques that do not fully exploit the label information

inherent in the data. To address this, the current subsection redefines the distance metric within the KUMAP framework, extending it into a supervised dimensionality reduction algorithm.

Notably, traditional categorical labels often fail to capture the underlying geometric relationships between different clusters. To overcome this limitation, we redefine the label vectors by incorporating spatial information. For a dataset $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ comprising C classes, PCA is first employed to reduce its dimensionality. The centroid of each class in this low-dimensional space is then designated as the new representative label for that class. The refined label for the i -th class, denoted as $\mathbf{t}_i \in \mathbb{R}^n$, is calculated as follows:

$$\mathbf{t}_i = \frac{1}{N_i} \sum_{j=1}^{N_i} \text{PCA}(\mathbf{x}_j^i) \quad i = 1, 2, 3, \dots, C \quad (12)$$

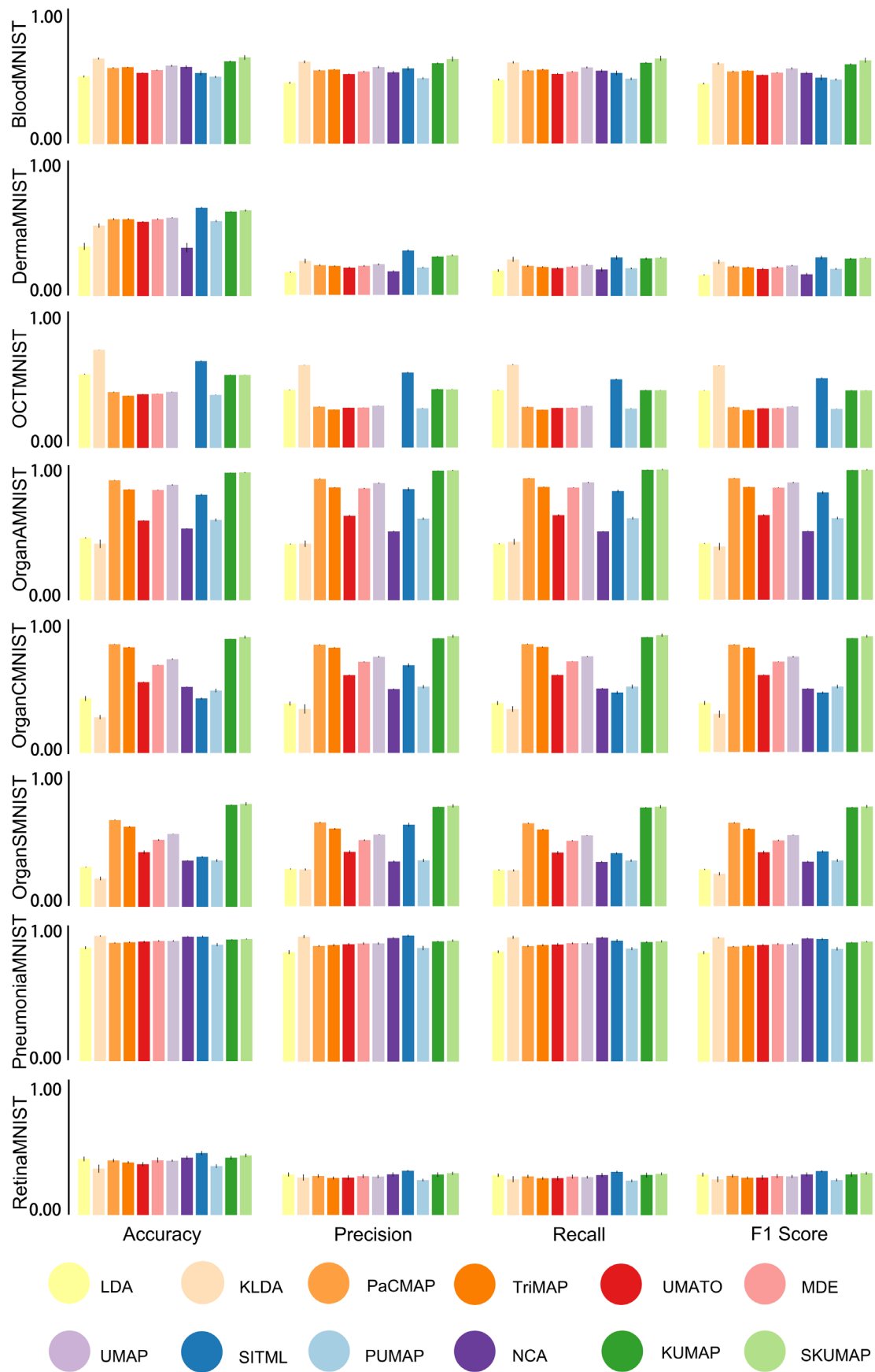


FIGURE 3 | Visualisation of the average classification performance of 12 algorithms across 8 datasets. In each subplot, the bar represents the average classification performance, with error bars indicating the standard deviation.

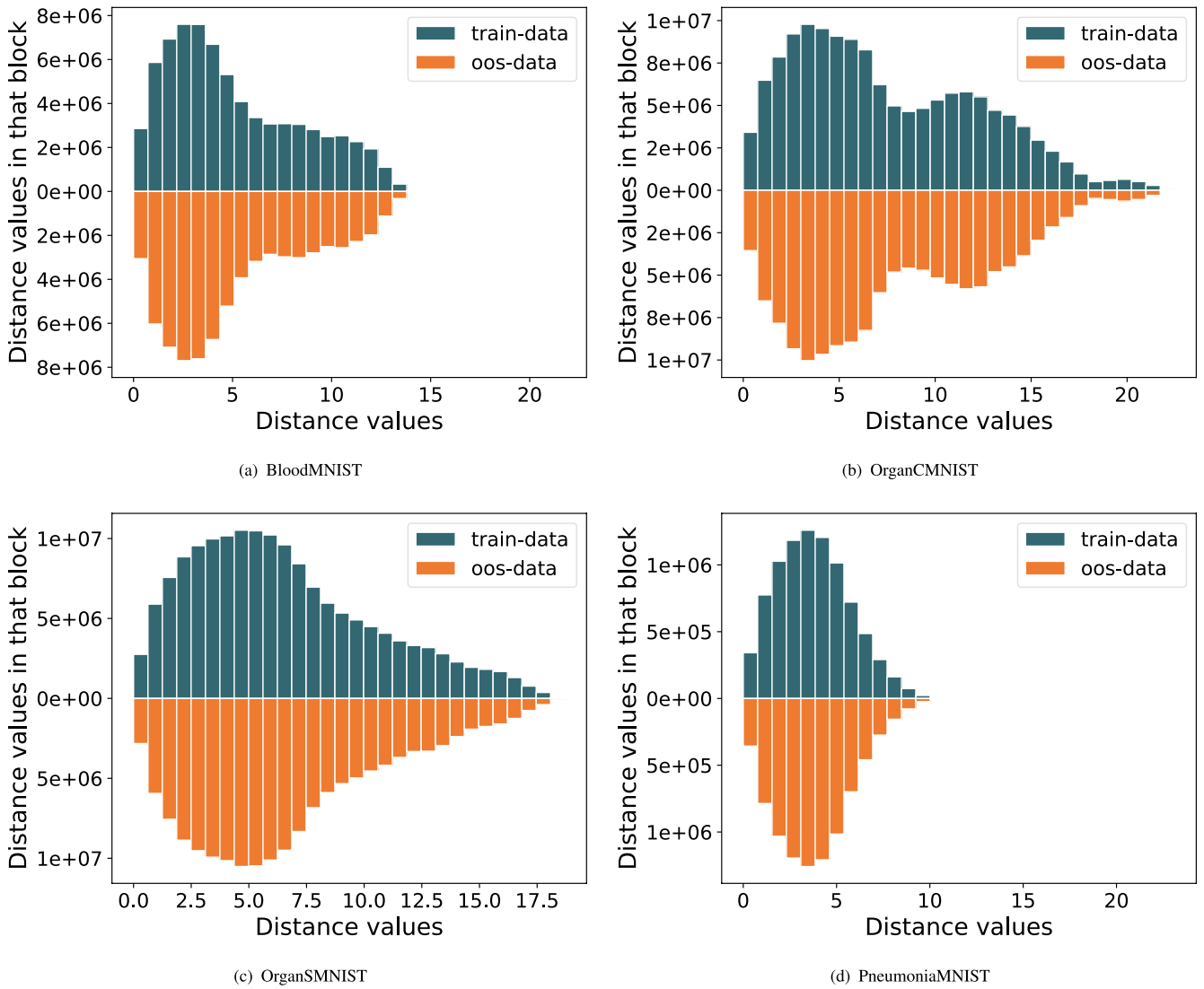


FIGURE 4 | Frequency histograms of distance matrices for data on the low-dimensional manifold generated by KUMAP. In each subplot, the region above the x-axis illustrates the distribution of training data on the manifold, while the region below the x-axis represents the distribution of out-of-sample points.

where N_i is the number of samples in the i -th class, $\mathbf{x}_i^j$ is the j th sample of that class and PCA denotes the PCA projection mapping. By integrating these label vectors, the distance between any two samples $\mathbf{x}_i$ and $\mathbf{x}_j$ in the proposed SKUMAP is redefined as:

$$d_{SKUMAP}(\mathbf{x}_i, \mathbf{x}_j) = \lambda \cdot \|\mathbf{x}_i - \mathbf{x}_j\| + (1 - \lambda) \cdot \|\mathbf{t}_{c(i)} - \mathbf{t}_{c(j)}\| \quad (13)$$

where $c(i)$ and $c(j)$ denote the class indices of $\mathbf{x}_i$ and $\mathbf{x}_j$, respectively. The parameter $\lambda \in (0, 1)$ acts as a weight balancing the contribution of the original feature space and the label space.

By incorporating this new metric d_{SKUMAP} , we introduce the SKUMAP algorithm, which builds upon and enhances the KUMAP foundation.

4 | Experiments

To rigorously evaluate the efficacy of the proposed KUMAP and SKUMAP frameworks, we conducted a series of comprehensive experiments focused on two core dimensions: classification performance on high-dimensional medical images and the robustness of Out-of-Sample embeddings. Benchmarks were performed across eight diverse medical image datasets (including BloodMNIST, DermaMNIST and OCTMNIST), representing a broad spectrum of diagnostic challenges. The experimental design is structured as follows:

- *Performance Benchmarking in Medical Classification:* We employed KUMAP and SKUMAP as feature extractors to project high-dimensional images onto a low-dimensional manifold. A KNN classifier was subsequently utilised to assess the discriminative power of the learned embeddings.

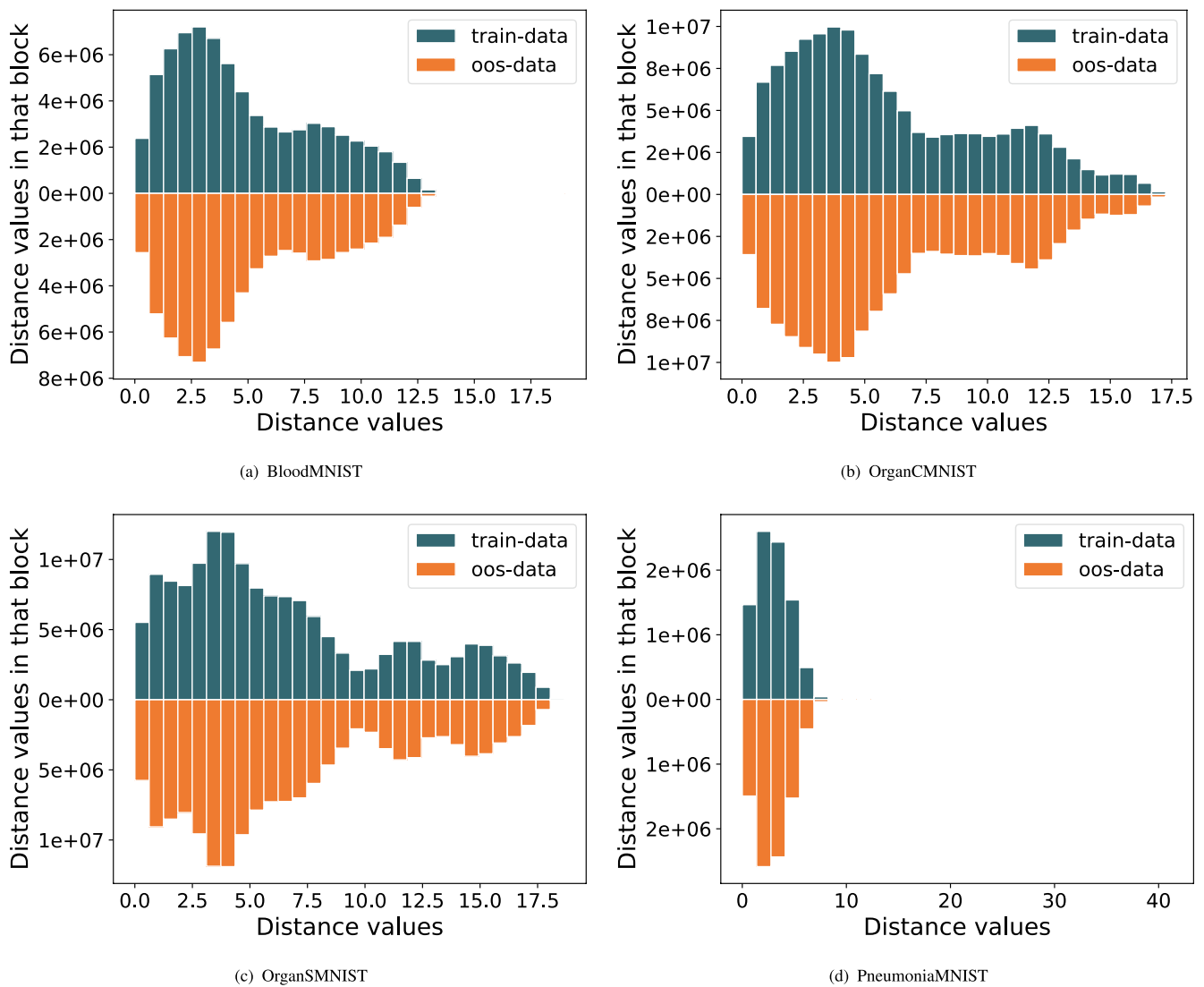


FIGURE 5 | Frequency histograms of distance matrices for data on the low-dimensional manifold generated by SKUMAP. In each subplot, the region above the x-axis illustrates the distribution of training data on the manifold, while the region below the x-axis represents the distribution of out-of-sample points.

This experiment aims to demonstrate that our kernelised approach preserves essential diagnostic features more effectively than standard UMAP and other state-of-the-art dimensionality reduction methods.

- **Validation of Out-of-Sample Embedding Capability:** To address the inherent limitations of UMAP in processing unseen points, we specifically evaluated the frameworks' ability to embed OOS samples without recomputing the entire manifold. By visualising the low-dimensional distribution of OOS samples and comparing them with the training set, we provide evidence of the frameworks' stability and generalisation performance.
- **Ablation and Comparative Analysis:** A comparative study was conducted between the unsupervised KUMAP and its supervised counterpart, SKUMAP. This analysis reveals how incorporating redefinition label information enhances class separability and mitigates the impact of noise in complex medical datasets.

- **Computational Efficiency Analysis:** Furthermore, we recorded the operational costs of KUMAP, SKUMAP and baseline methods, including execution time and memory usage.
- **Sensitivity Analysis:** Finally, a sensitivity analysis was performed on key hyperparameters influencing performance, such as the training sample proportion, the dimensionality of custom label vectors η and the data information proportion λ .

4.1 | Datasets

To evaluate the performance of KUMAP and SKUMAP in medical image classification tasks, experiments were conducted on eight 2D medical image datasets sourced from MedMNIST-v2^{1,2} (Yang et al. 2023). The key characteristics of these datasets, including the number of samples, classes and dimensions, are summarised in Table 1.

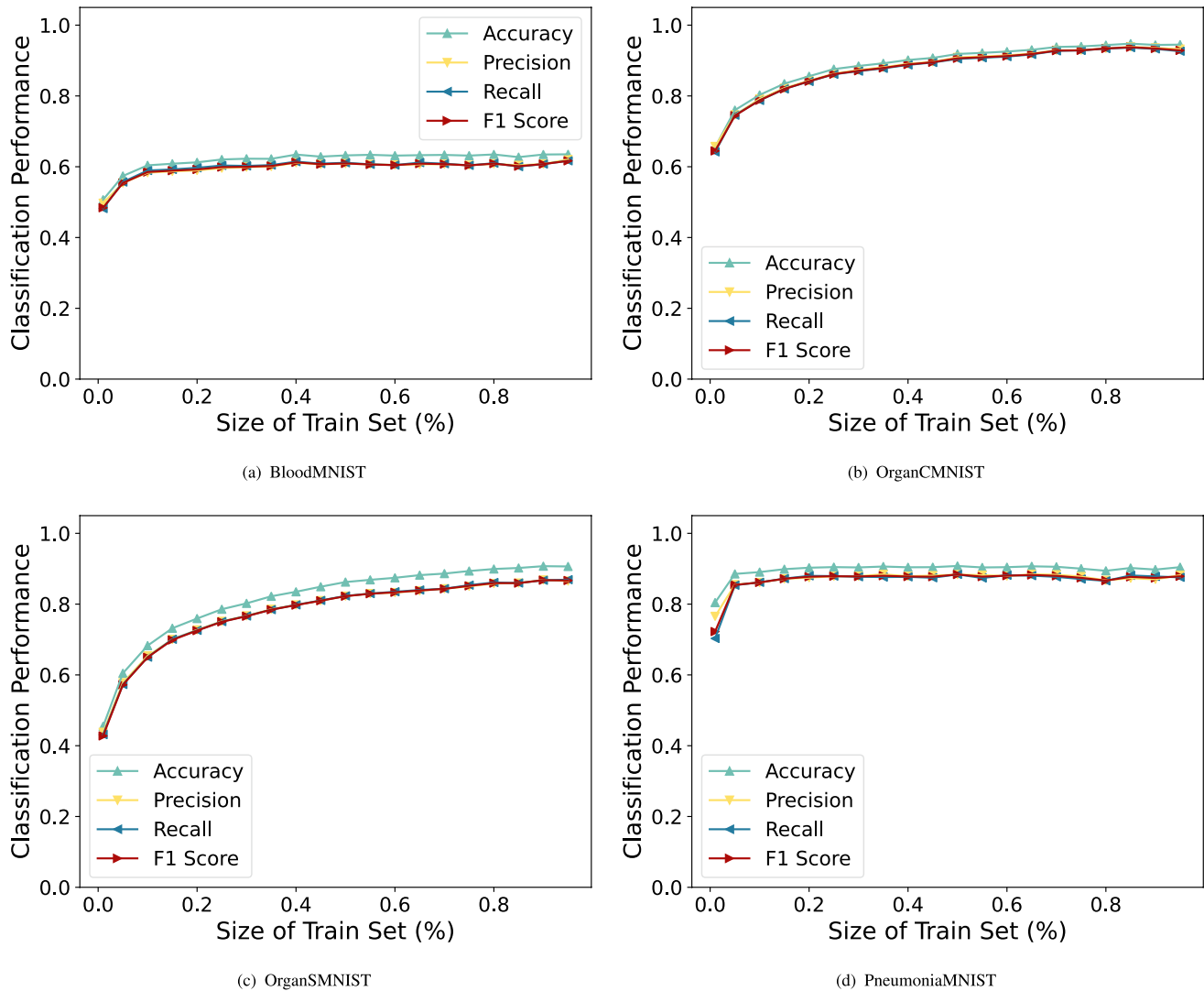


FIGURE 6 | Classification performance of KUMAP under varying size of train set across four datasets.

4.2 | Experimental Details

To evaluate the performance of KUMAP and SKUMAP on medical image classification tasks, we conducted comparative experiments with several representative dimensionality reduction methods across eight 2D datasets from MedMNIST-v2 (Yang et al. 2023): BloodMNIST, DermaMNIST, OCTMNIST, OrganAMNIST, OrganCMNIST, OrganSMNIST, PneumoniaMNIST and RetinaMNIST. The baseline methods are categorised as follows:

1. Discriminant Dimensionality Reduction

- *LDA*³ (Linear Discriminant Analysis) (Fisher 1936): Identifies the optimal linear projection by maximising the ratio of between-class scatter to within-class scatter.
- *KLDA*⁴ (Kernel LDA) (Mika et al. 1999; Baudat and Anouar 2000): Leverages the kernel trick to map data into a high-dimensional feature space, enabling LDA to capture nonlinear discriminative structures that are not linearly separable in the original space.

2. Supervised Metric Learning

- *NCA*⁵ (Neighbourhood Components Analysis) (Goldberger et al. 2005): Maximises the probability that each sample is correctly classified by its nearest neighbours in a leave-one-out sense through optimising the Mahalanobis distance.
- *SITML*⁶ (Supervised Information Theoretic Metric Learning) (Davis et al. 2007): Learns a metric matrix by minimising the Kullback–Leibler (KL) divergence between two multivariate Gaussian distributions, subject to constraints on the upper bounds of distances between similar pairs and lower bounds for dissimilar pairs.

3. State-of-the-Art Manifold Learning

- *PaCMAP*⁷ (Pairwise Controlled Manifold Approximation Projection) (Wang et al. 2021): Simultaneously preserves global topology and local neighbourhoods by balancing attractive and repulsive forces across near, mid-range and distant sample pairs.
- *TriMAP*⁸ (Amid and Warmuth 2019): Preserves both local structure and global topology in the low-dimensional

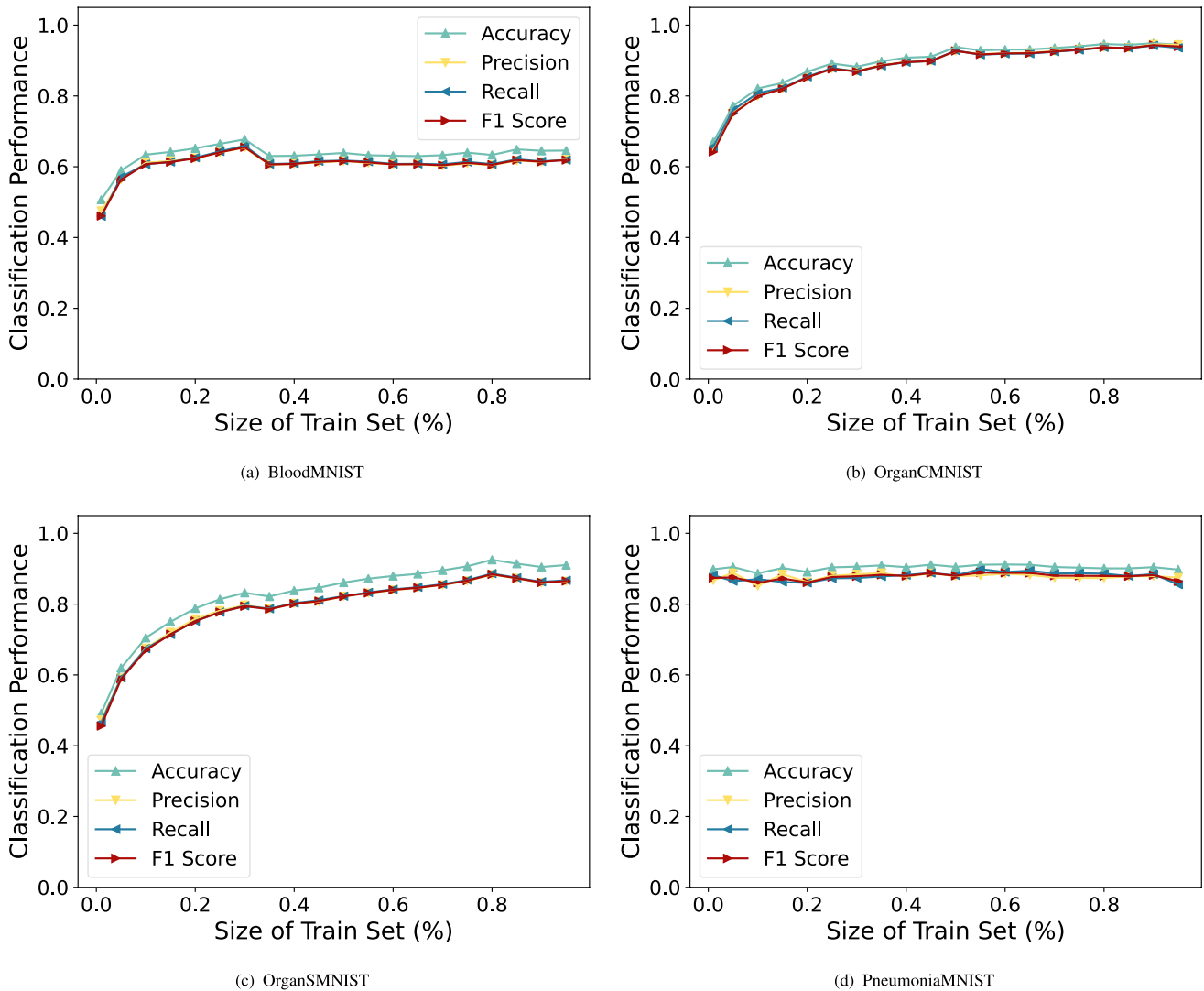


FIGURE 7 | Classification performance of SKUMAP under varying size of train set across four datasets.

space by maintaining the relative distance relationships within sampled triplets.

- *UMATO*⁹ (Uniform Manifold Approximation with Two-phase Optimisation) (Jeon et al. 2022): Introduces a two-step optimisation strategy within the UMAP framework, prioritising the embedding of global hub points to construct a topological skeleton, thereby improving embedding quality and global stability.
- *MDE*¹⁰ (Minimum Distortion Embedding) (Agrawal et al. 2021): Formulates dimensionality reduction as a distortion minimisation problem, utilising custom distance functions and Laplacian constraints to achieve flexible low-dimensional projections.

4. OOS Methods for UMAP

- *PUMAP*¹¹ (Parametric UMAP) (Leland et al. 2018; Sainburg et al. 2021): Employs a deep neural network as a parametric model to approximate the UMAP projection function, thereby enabling efficient inference for novel samples and enhancing scalability to large-scale datasets.

For methods with explicit projection capabilities, specifically LDA, KLDA, NCA, SITML, PUMAP and our proposed KUMAP

and SKUMAP, novel samples are directly mapped to the low-dimensional space. To evaluate these methods, we utilised 20% of the data as a training set to train both the feature extraction models and a K-Nearest Neighbours (KNN) classifier; the remaining 80% was reserved as OOS points for performance evaluation.

In contrast, non-linear methods such as UMAP, PaCMAP, TriMAP, UMATO and MDE lack inherent OOS capabilities and cannot directly embed unseen samples. To assess their performance, we first performed dimensionality reduction on the entire dataset. The resulting low-dimensional embeddings were then split into training and testing sets at a 2:8 ratio. The classification efficacy of these manifold learning methods was subsequently evaluated using a KNN classifier. This transductive evaluation ensures that the intrinsic representation quality of these non-linear methods can be benchmarked, despite their inability to handle out-of-sample data directly.

Throughout this study, global parameters, such as the training sample ratio, target dimensionality and classifier settings, were maintained consistent across all models. For method-specific hyperparameters, we adhered to the default configurations provided in their respective open-source implementations.

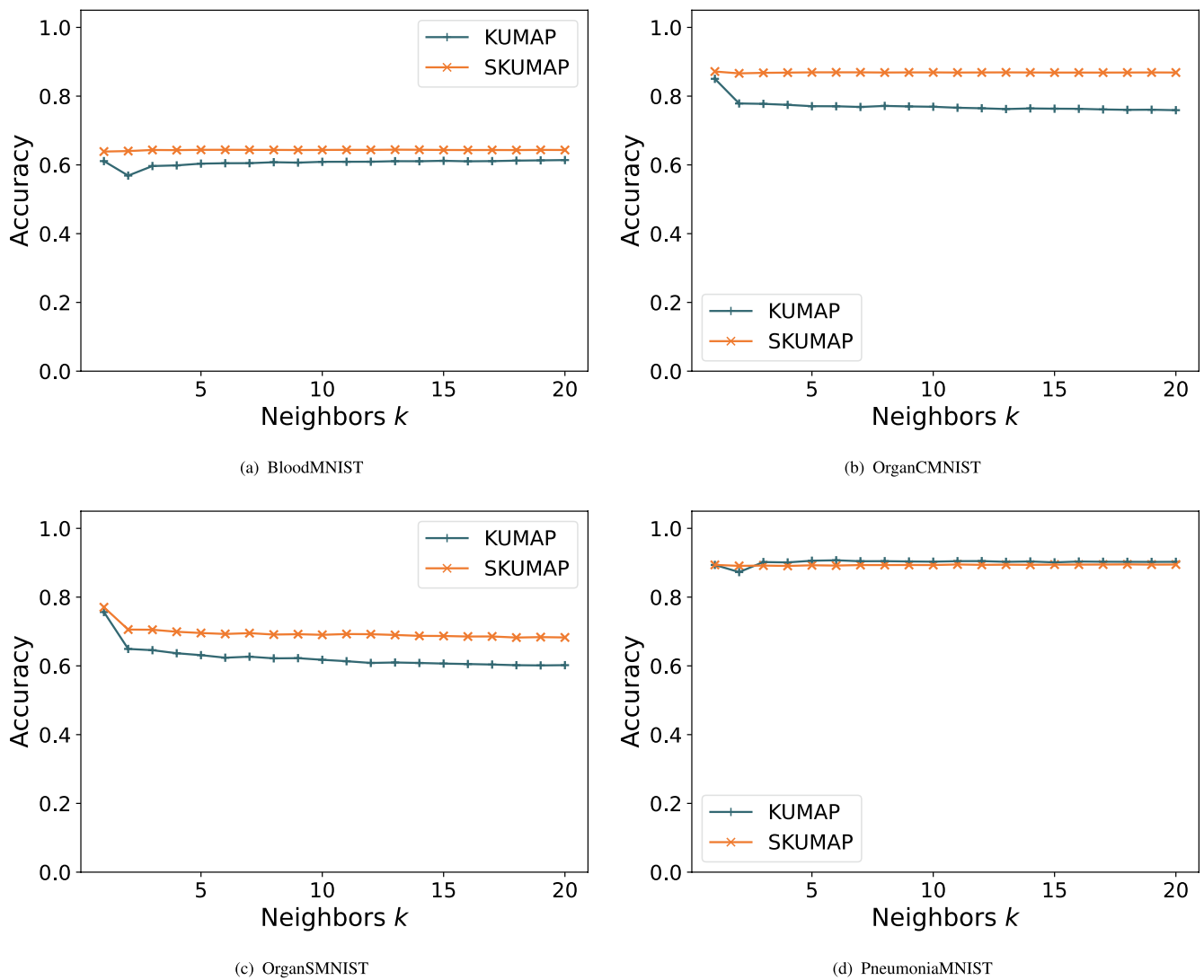


FIGURE 8 | Classification accuracy of KUMAP and SKUMAP with varying neighbourhood sizes k across four datasets.

Detailed parameter settings for each method are summarised in Table 2. To minimise the impact of stochastic variation, each experiment was executed independently 10 times. The feature extraction efficacy was quantified using the mean classification accuracy, precision, recall and $F1$ -score of the KNN classifier.

4.3 | Experimental Results

To evaluate the effectiveness of the proposed KUMAP and SKUMAP algorithms in medical image classification, extensive comparative experiments were conducted across 8 datasets. The average classification accuracy, precision, recall and $F1$ -score of these methods are reported in Tables 3–6 and visualised in Figure 3.

Based on the experimental results of the 12 algorithms across 8 medical image datasets presented in Tables 3–6, the significant advantages and clinical efficacy of the proposed KUMAP and SKUMAP algorithms in medical image classification tasks can be systematically demonstrated.

In terms of Accuracy, KUMAP and SKUMAP achieved optimal or sub-optimal performance across most tasks in the eight datasets, particularly on complex multi-class datasets such as OrganAMNIST, OrganCMNIST and OrganSMNIST. This indicates that the proposed methods effectively enhance class separability while maintaining the global structure, thereby improving overall diagnostic accuracy. Furthermore, the generally small standard deviations of both methods across different datasets demonstrate their excellent stability and robustness, which are critical for generalisation across data from diverse sources in medical scenarios.

Regarding Precision, KUMAP and SKUMAP achieved the highest or near-highest performance on multiple datasets, with a distinct advantage in the BloodMNIST and Organ series. In medical diagnosis, precision is directly related to the misdiagnosis rate; higher precision means the model can effectively reduce the misclassification of healthy samples as pathological, thus avoiding unnecessary further examinations or treatments. In contrast, some baselines (e.g., LDA, PUMAP) exhibited significantly lower precision, indicating substantial errors at the decision boundaries. By contrast, KUMAP and SKUMAP effectively

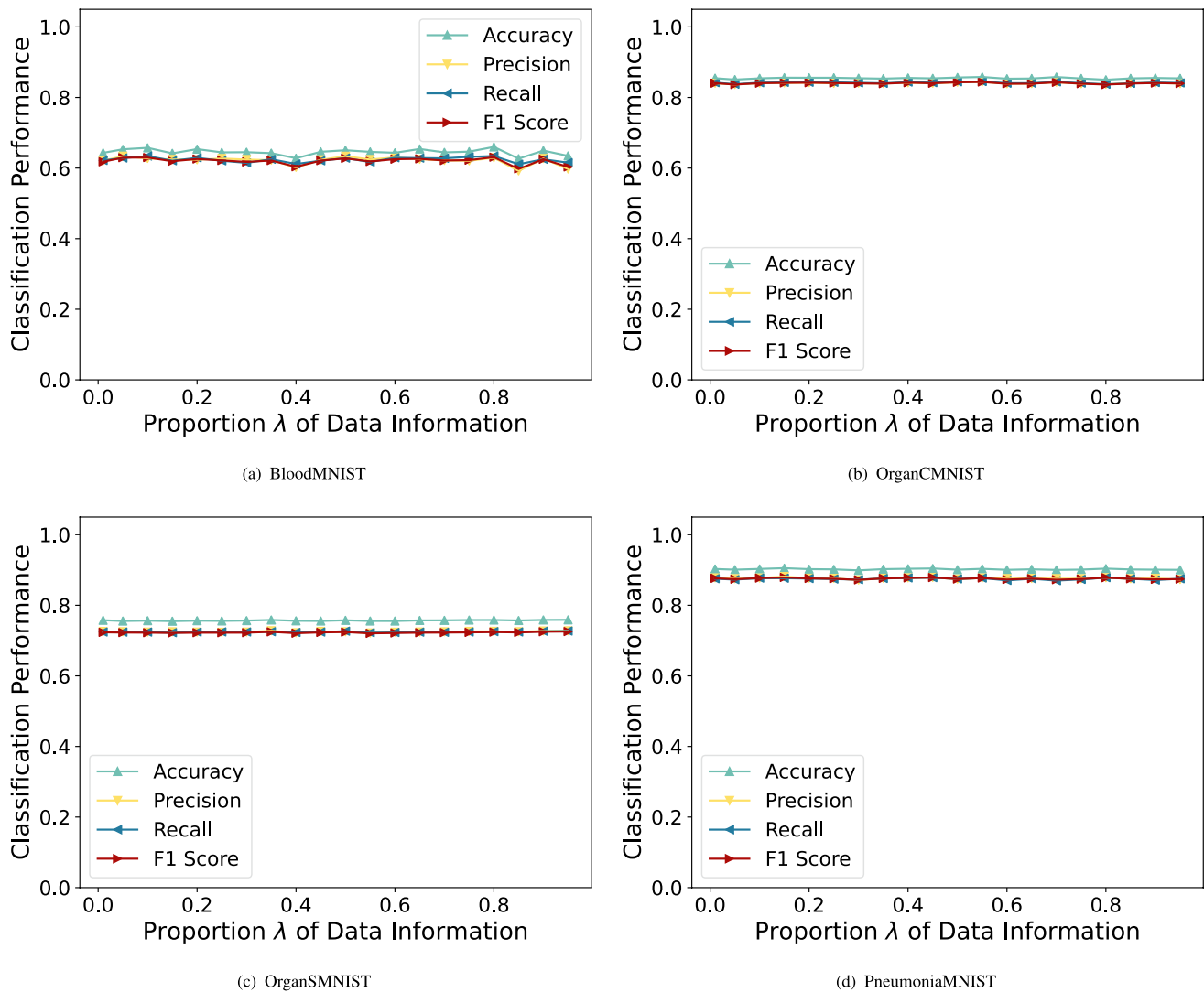


FIGURE 9 | Classification performance of SKUMAP with varying data information proportion λ across four datasets.

improve inter-class separation in the feature space through kernelised modelling and structural optimisation.

In terms of Recall, KUMAP and SKUMAP also demonstrate significant advantages, achieving optimal or sub-optimal results on datasets like BloodMNIST, OrganCMNIST and OrganSMNIST. Recall reflects the model's ability to identify positive samples, corresponding to the risk of missed diagnosis in computer-aided medical diagnosis. Higher recall implies that the model can identify as many potential lesion samples as possible, which is crucial for early screening and disease prevention. While KLDA showed high recall on certain datasets (e.g., PneumoniaMNIST), its performance fluctuated significantly across others, lacking stability. In comparison, KUMAP and SKUMAP maintained consistently high recall across multiple datasets.

For *F1*-score, KUMAP and SKUMAP achieved the best or second-best results on most datasets, reflecting a superior balance between precision and recall. As a comprehensive metric, the *F1*-score fully reflects the model's ability to trade

off between reducing misdiagnosis and missed diagnosis. In clinical practice, optimising a single metric often leads to the degradation of another; however, KUMAP and SKUMAP maintain both high precision and recall simultaneously, achieving superior overall diagnostic performance.

In summary, KUMAP and SKUMAP demonstrate significant advantages across four key metrics: accuracy, precision, recall and *F1*-score, showing stable and superior performance especially on complex multi-class medical image datasets. This not only illustrates that the proposed methods possess a stronger capability in manifold structure preservation but also signifies their practical value in lowering misdiagnosis risks, reducing missed diagnosis probabilities, and achieving balanced diagnostic performance. The overall performance of SKUMAP outperforms KUMAP, suggesting that its utilisation of structural constraints or supervisory information further enhances the model's discriminative ability. Therefore, KUMAP and SKUMAP provide an effective and reliable dimensionality reduction and feature learning solution for computer-aided medical diagnosis.

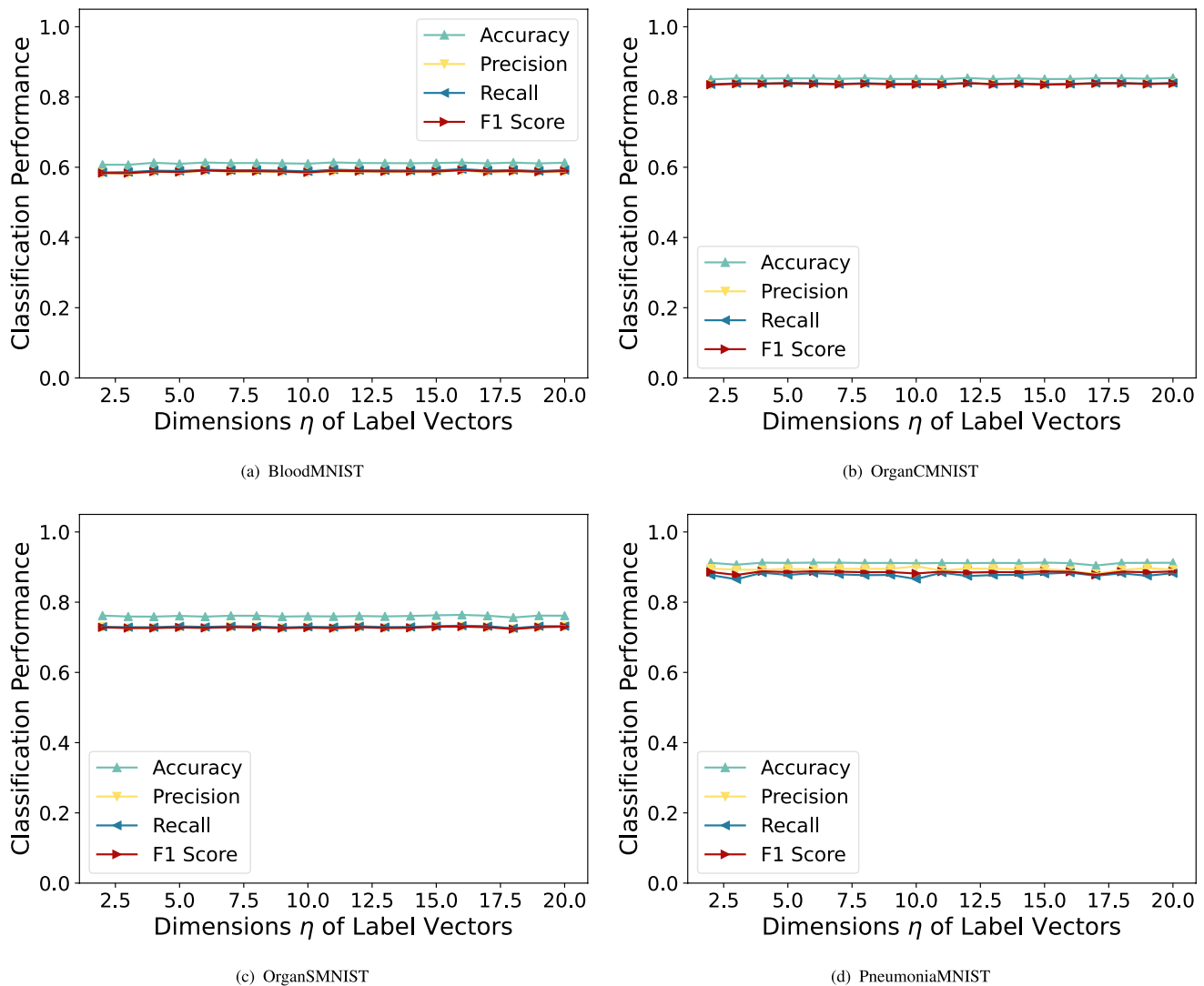


FIGURE 10 | Classification performance of SKUMAP with varying label vectors dimensions η across four datasets.

4.4 | The Generalisation of KUMAP and SKUMAP

As for the KUMAP and SKUMAP algorithms, another question worth investigating is the fitting performance of the coefficient matrix $\mathbf{A}$ on out-of-sample points. To explore this issue, we evenly split the BloodMNIST, OrganCMNIST, OrganSMNIST and PneumoniaMNIST datasets into training and out-of-sample sets. Furthermore, we visualised the distribution of distances between projected points on the low-dimensional manifold. In Figures 4 and 5, the region above the x -axis represents the distribution of training data projections, while the region below the x -axis illustrates the distribution of out-of-sample points projections.

As intuitively observed from the charts, the distributions of the training data (above the x -axis) and the out-of-sample points (below the x -axis) exhibit an exceptionally high degree of symmetry. This implies that the coefficient matrix $\mathbf{A}$, while capturing the manifold structure of the training set, can accurately map unseen samples to consistent positions in the low-dimensional space. Such distributional consistency systematically demonstrates that both KUMAP and SKUMAP possess

superior generalisation performance and can effectively mitigate overfitting.

Regarding the smaller-scale PneumoniaMNIST dataset shown in Figures 4d and 5d, the projected distance distribution maintains rigorous mirror symmetry. This indicates that the proposed dimensionality reduction scheme exhibits robust stability across the diverse data distributions characteristic of medical images. Consequently, the scheme provides reliable and consistent feature inputs for downstream clinical classification tasks, further validating its potential efficacy as a computer-aided diagnostic tool.

4.5 | Parameters Analysis

This subsection performs a rigorous sensitivity analysis on several key hyperparameters that influence the performance of the KUMAP and SKUMAP algorithms. To ensure the generalizability of the findings, we evaluate the proposed methods on four representative medical image benchmarks: BloodMNIST, OrganCMNIST, OrganSMNIST and PneumoniaMNIST.

TABLE 7 | Average execution time (s $\pm$ standard deviation) of 12 algorithms across 8 datasets.

Methods	BloodMNIST	DermaMNIST	OCTMNIST	OrganAMNIST
LDA	182.0484 $\pm$ 1.8631	135.1523 $\pm$ 3.5949	41.9862 $\pm$ 1.2591	<u>37.8724 $\pm$ 0.9538</u>
KLDA	42.8182 $\pm$ 1.3300	33.1294 $\pm$ 0.4867	457.3568 $\pm$ 5.9132	144.1404 $\pm$ 1.1266
PaCMAP	<u>53.0167 $\pm$ 2.3672</u>	32.4491 $\pm$ 2.0401	343.4858 $\pm$ 44.6087	151.8117 $\pm$ 7.8228
TriMAP	75.4703 $\pm$ 4.2133	46.6760 $\pm$ 2.0036	572.1415 $\pm$ 89.0386	251.0624 $\pm$ 20.6722
UMATO	214.3254 $\pm$ 9.1836	<u>31.9606 $\pm$ 1.6774</u>	200.9186 $\pm$ 18.8320	106.1784 $\pm$ 8.3571
MDE ^a	18.1372 $\pm$ 1.8146	5.5346 $\pm$ 0.9460	24.2131 $\pm$ 2.6971	18.0793 $\pm$ 2.6921
UMAP	136.3568 $\pm$ 4.3153	32.2182 $\pm$ 1.8451	222.0430 $\pm$ 12.3894	259.9418 $\pm$ 21.8519
NCA	418.3323 $\pm$ 34.3104	220.8667 $\pm$ 9.1315	—	2939.1604 $\pm$ 452.2999
SITML	188.9814 $\pm$ 22.5258	1102.0113 $\pm$ 179.7060	<u>172.6727 $\pm$ 18.7015</u>	27.1313 $\pm$ 2.2070
PUMAP	523.4966 $\pm$ 34.6814	257.0781 $\pm$ 18.2481	797.0309 $\pm$ 49.0304	390.2606 $\pm$ 29.5091
KUMAP	96.0959 $\pm$ 5.6350	30.8876 $\pm$ 2.6559	280.8279 $\pm$ 38.1691	97.4883 $\pm$ 19.8858
SKUMAP	122.5264 $\pm$ 4.2270	36.9245 $\pm$ 5.9617	310.9340 $\pm$ 40.5789	100.8005 $\pm$ 17.6407
Methods	OrganCMNIST	OrganSMNIST	PneumoniaMNIST	RetinaMNIST
LDA	33.5520 $\pm$ 1.2873	33.5131 $\pm$ 1.1712	29.4830 $\pm$ 1.3641	24.1239 $\pm$ 1.0415
KLDA	51.7610 $\pm$ 0.8771	54.9054 $\pm$ 1.0819	25.0870 $\pm$ 0.6745	0.6845 $\pm$ 0.0486
PaCMAP	64.1468 $\pm$ 2.4686	68.0970 $\pm$ 2.6610	21.0731 $\pm$ 0.7934	<u>12.1197 $\pm$ 0.5520</u>
TriMAP	98.3450 $\pm$ 4.3165	103.9062 $\pm$ 5.1204	29.5425 $\pm$ 1.1081	14.5000 $\pm$ 0.5328
UMATO	47.8096 $\pm$ 2.9308	51.5888 $\pm$ 3.0604	21.0722 $\pm$ 0.9926	14.2659 $\pm$ 0.6527
MDE ^a	9.0284 $\pm$ 0.7069	10.3688 $\pm$ 1.5249	3.7352 $\pm$ 0.3599	4.2697 $\pm$ 0.1933
UMAP	58.5733 $\pm$ 2.1567	62.1940 $\pm$ 3.3367	20.6254 $\pm$ 1.3389	47.8448 $\pm$ 3.2582
NCA	545.6571 $\pm$ 75.2378	588.3113 $\pm$ 71.8273	15.2664 $\pm$ 1.1703	30.2163 $\pm$ 1.4988
SITML	26.3091 $\pm$ 2.6339	27.5090 $\pm$ 3.5980	155.6204 $\pm$ 22.9627	2902.9669 $\pm$ 767.5313
PUMAP	408.4750 $\pm$ 30.5711	440.5253 $\pm$ 31.1034	108.5832 $\pm$ 9.0326	67.2332 $\pm$ 4.6459
KUMAP	<u>29.8713 $\pm$ 3.5136</u>	<u>33.0066 $\pm$ 6.4652</u>	<u>19.2514 $\pm$ 1.2686</u>	13.0462 $\pm$ 0.7490
SKUMAP	40.7508 $\pm$ 6.1095	39.3297 $\pm$ 6.1574	22.6635 $\pm$ 2.4393	12.3466 $\pm$ 0.6184

Note: The best and second-best results (i.e., the lowest values) are highlighted in bold and underlined, respectively.

^aTo ensure a fair comparison, MDE was excluded from the performance ranking because it was executed with GPU acceleration on an RTX 2080 Ti.

4.5.1 | Impact of Training Data Proportion

We systematically investigated the influence of the proportion of the training set on the classification performance of KUMAP and SKUMAP over a comprehensive interval of [0.01,0.95]. The experimental results, visualised in Figures 6 and 7, reveal a consistent trend across all datasets.

Specifically, the classification performance exhibits a sharp upward trajectory in the low-proportion regime (below 10%) and begins to converge as the proportion reaches approximately 20%. This rapid stabilisation indicates that the proposed kernelised manifold learning framework possesses high sample efficiency, requiring only a small subset of the total data to learn a robust and representative coefficient matrix **A** for out-of-sample projection. Beyond the 20% threshold, additional training data yields marginal gains in performance, suggesting that the

fundamental manifold structure has been adequately captured. Consequently, to balance computational efficiency with model performance, we fixed the training data proportion at 20% for all subsequent comparative experiments.

4.5.2 | Sensitivity Analysis of the Classifier to Neighbourhood Size

To evaluate the robustness of the learned low-dimensional embeddings, we examined the impact of the neighbourhood size k in the KNN classifier, varying k from 1 to 20. The resulting accuracy curves are illustrated in Figure 8.

Our empirical results demonstrate that both KUMAP and SKUMAP achieve optimal or near-optimal performance at $k = 1$ across the majority of datasets, particularly on the complex

TABLE 8 | Average memory consumption (MB $\pm$ standard deviation) of 12 algorithms across 8 datasets.

Methods	BloodMNIST	DermaMNIST	OCTMNIST	OrganAMNIST
LDA	729.8132 $\pm$ 0.2896	446.8678 $\pm$ 0.1016	1262.7497 $\pm$ 0.3588	<u>664.4434 $\pm$ 0.1310</u>
KLDA	665.5697 $\pm$ 0.0003	303.1809 $\pm$ 0.0001	15248.3065 $\pm$ 0.0003	4583.3685 $\pm$ 0.0002
PaCMAP	460.1017 $\pm$ 0.0001	269.5777 $\pm$ 0.0000	980.7476 $\pm$ 0.0000	527.8416 $\pm$ 0.0000
TriMAP	628.2216 $\pm$ 0.0002	368.5005 $\pm$ 0.0000	1395.6002 $\pm$ 0.0000	751.2874 $\pm$ 0.0000
UMATO	1386.5484 $\pm$ 0.4609	1209.6110 $\pm$ 0.1421	1899.0947 $\pm$ 0.1357	1484.2644 $\pm$ 0.3469
MDE ^a	153.4113 $\pm$ 0.0001	89.8650 $\pm$ 0.0000	326.9217 $\pm$ 0.0000	175.9530 $\pm$ 0.0000
UMAP	1426.2980 $\pm$ 0.3343	1224.2539 $\pm$ 0.2936	2010.9502 $\pm$ 0.5580	1536.9812 $\pm$ 0.2466
NCA	954.8198 $\pm$ 0.0029	446.8848 $\pm$ 0.0677	—	8010.8031 $\pm$ 0.0008
SITML	655.9594 $\pm$ 0.0002	401.8267 $\pm$ 0.0002	1313.2495 $\pm$ 0.0002	708.9889 $\pm$ 0.0001
PUMAP	1166.9227 $\pm$ 1.1241	639.6995 $\pm$ 0.6320	2560.1058 $\pm$ 1.9876	1457.0137 $\pm$ 0.1678
KUMAP	<u>180.4488 $\pm$ 0.0017</u>	61.7207 $\pm$ 0.0004	1082.2335 $\pm$ 0.8386	1025.1751 $\pm$ 0.0248
SKUMAP	180.4485 $\pm$ 0.0011	<u>64.0791 $\pm$ 0.0001</u>	<u>1073.8657 $\pm$ 0.4346</u>	1032.4086 $\pm$ 0.0240
Methods	OrganCMNIST	OrganSMNIST	PneumoniaMNIST	RetinaMNIST
LDA	<u>280.9571 $\pm$ 0.0823</u>	<u>298.4730 $\pm$ 0.0990</u>	87.0375 $\pm$ 0.0709	58.5657 $\pm$ 0.0397
KLDA	822.4757 $\pm$ 0.0001	929.4766 $\pm$ 0.0002	77.3960 $\pm$ 0.0002	32.0083 $\pm$ 0.0002
PaCMAP	211.5996 $\pm$ 0.0000	226.2062 $\pm$ 0.0000	52.5499 $\pm$ 0.0000	43.0753 $\pm$ 0.0000
TriMAP	301.3955 $\pm$ 0.0000	322.1753 $\pm$ 0.0000	75.1285 $\pm$ 0.0000	59.7562 $\pm$ 0.0000
UMATO	1195.2736 $\pm$ 0.1247	1208.2049 $\pm$ 0.1609	1052.8150 $\pm$ 0.0395	1039.4902 $\pm$ 0.0448
MDE ^a	70.5390 $\pm$ 0.0000	75.4079 $\pm$ 0.0000	18.2375 $\pm$ 0.0000	14.4621 $\pm$ 0.0000
UMAP	1212.2088 $\pm$ 0.2902	1226.3738 $\pm$ 0.2924	1054.4179 $\pm$ 0.0599	68.9116 $\pm$ 0.0038
NCA	1372.1570 $\pm$ 0.0020	1557.8813 $\pm$ 0.0021	111.5280 $\pm$ 0.0015	58.5091 $\pm$ 0.0389
SITML	287.0639 $\pm$ 0.0003	306.5519 $\pm$ 0.0002	74.8625 $\pm$ 0.0002	212.9422 $\pm$ 0.0683
PUMAP	1186.9818 $\pm$ 0.0880	1218.5010 $\pm$ 1.0188	280.6561 $\pm$ 0.7559	94.2445 $\pm$ 0.1939
KUMAP	1014.5245 $\pm$ 0.0131	1015.0982 $\pm$ 0.0185	21.2832 $\pm$ 0.0589	10.3649 $\pm$ 0.2330
SKUMAP	1017.2959 $\pm$ 0.0160	1018.1701 $\pm$ 0.0109	<u>31.0171 $\pm$ 0.0001</u>	<u>11.1700 $\pm$ 0.1762</u>

Note: The best and second-best results (i.e., the lowest consumption) are highlighted in bold and underlined, respectively.

^aTo ensure a fair comparison, MDE was excluded from the performance ranking because it was executed with GPU acceleration on an RTX 2080 Ti.

Organ-series (OrganCMNIST and OrganSMNIST). Notably, as k increases, the classification accuracy remains remarkably stable with negligible fluctuations. This phenomenon suggests that our algorithms effectively enhance inter-class separability and intra-class compactness in the projected manifold space. The high discriminative ability of the features ensures that a simple nearest-neighbour rule is sufficient for accurate classification, further proving the reliability of the learned features. Based on these observations, we consistently set $k = 1$ for the KNN classifiers in the benchmarking phase to minimise hyper-parameter complexity.

4.5.3 | Sensitivity Analysis of the Proportion of Data Information

This experiment aims to investigate the impact of the label information proportion parameter λ , on the classification

performance of the SKUMAP model, thereby evaluating the robustness of the algorithm under varying degrees of supervisory guidance. In the experimental setup, the value of λ is specified within the range of [0.01,0.95]. Through incremental adjustments, the variations in the model's performance across various medical image datasets are observed, with the corresponding experimental results presented in Figure 9.

Based on the experimental results in Figure 9, SKUMAP demonstrates exceptional stability across all datasets. Whether in PneumoniaMNIST, where classification difficulty is lower (with all metrics remaining stable above 80%), or in BloodMNIST, which presents a higher challenge (with metrics stabilising around 60%), the four key indicators: Accuracy, Precision, Recall and $F1$ Score, show no significant fluctuations as the value of λ increases. The overall trend of the curves remains flat. This indicates that the SKUMAP model possesses

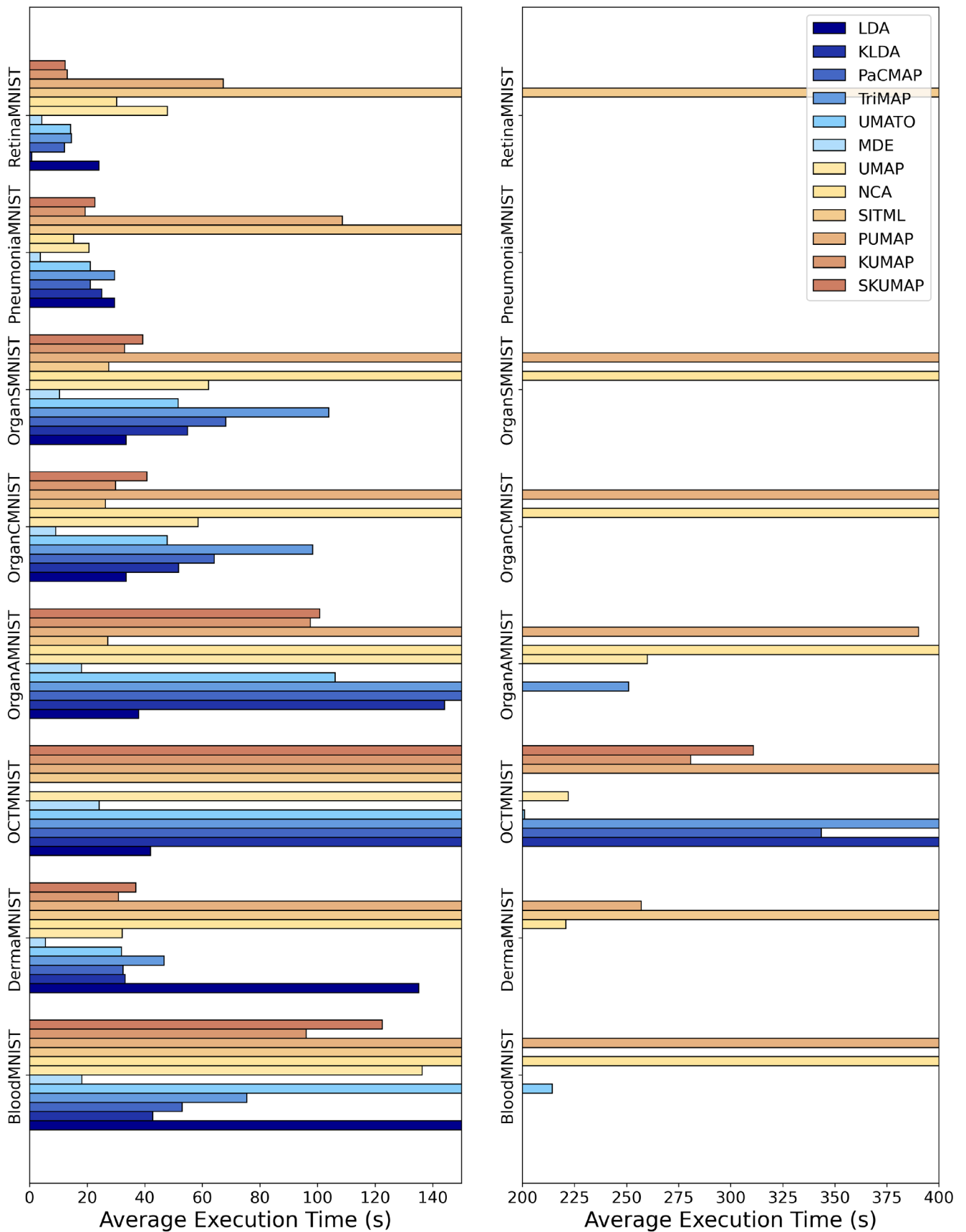


FIGURE 11 | Visualisation of the average execution time (s) of 12 algorithms across 8 datasets.

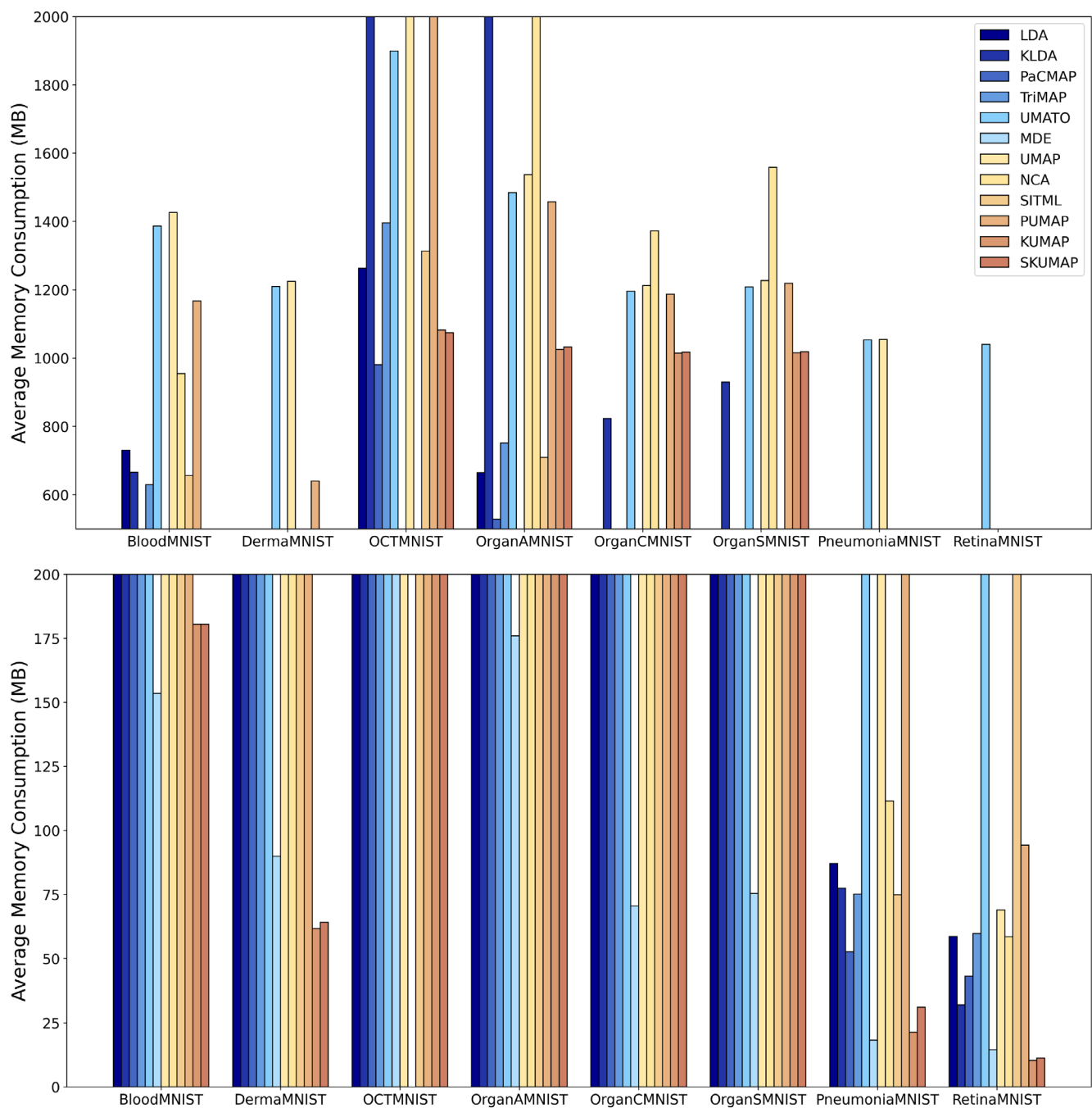


FIGURE 12 | Visualisation of the average memory consumption (MB) of 12 algorithms across 8 datasets.

excellent robustness, as its classification performance does not strictly depend on a specific weight distribution of label information, maintaining stable manifold embedding effects across different proportions of supervised information. In other experimental sections and the standard configuration of this study, the default value of λ is uniformly set to 0.2 to balance computational efficiency and classification performance. This setting introduces sufficient supervisory guidance to optimise the distribution in the feature space while avoiding potential overfitting caused by excessive reliance on labels; experiments have proven that this default parameter maintains universality across various tasks.

4.5.4 | Sensitivity Analysis of Label Vector Dimensions

This experiment aims to evaluate the robustness and classification stability of the SKUMAP algorithm under various dimensions of customised label vectors. The dimension of the label vector η is a critical hyperparameter that balances information representation capability with computational complexity. To investigate the quantitative impact of this parameter on classification performance, a search was conducted by varying the label vector dimension η within the range of [2, 20]. The model performance across different feature space dimensions was verified by observing the trends in four core metrics: Accuracy, Precision,

Recall and *F1* Score. The experimental results are visualised in Figure 10.

The experimental results demonstrate that SKUMAP exhibits exceptional dimensional insensitivity across all datasets. As observed in Figure 10, as the label vector dimension η increases from 2 to 20, the curves for all classification performance metrics remain stable, with no significant fluctuations or performance degradation. Specifically, on the PneumoniaMNIST dataset, performance consistently maintains a high level above 85%. Conversely, on BloodMNIST, although the overall performance baseline is lower (approximately 60%), the curves remain horizontal. This indicates that changes in the label vector dimension do not alter the guiding effectiveness of label information during the dimensionality reduction process. Based on this stability, and to balance computational efficiency with the richness of feature representation, this study sets the default dimension of the label vector to 8 for subsequent experiments. This configuration ensures that classification performance reaches saturation while effectively controlling the computational overhead associated with high-dimensional mapping.

4.6 | Experimental Cost Analysis

For evaluating dimensionality reduction and manifold learning algorithms, computational efficiency and memory overhead are critical metrics for assessing practical utility, alongside the capacity to preserve the topological structure of high-dimensional data. Particularly when processing large-scale datasets, the ability of algorithms to converge rapidly under constrained hardware resources directly dictates its deployment value in real-world scientific research and clinical settings. This subsection aims to reveal the resource consumption characteristics of various algorithms by quantitatively analysing the execution time and CPU memory usage of KUMAP, SKUMAP, and 10 baseline methods across eight representative datasets. Notably, while MDE is accelerated by a GPU (RTX 2080Ti), all other algorithms are executed on a 64-core CPU environment; thus, MDE results serve only as a reference for heterogeneous computing and are excluded from the CPU-based performance ranking.

Based on the experimental results in Tables 7 and 8, as well as the visualisations in Figures 11 and 12, a balanced analysis can be conducted from the following two dimensions:

4.6.1 | Computational Efficiency Analysis

Regarding computational efficiency, KUMAP demonstrates strong competitiveness, achieving the best performance on DermaMNIST and consistently ranking second on datasets such as OrganCMNIST and PneumoniaMNIST. In contrast, traditional methods like PUMAP and NCA exhibit significant latency when processing large-scale datasets (e.g., OrganAMNIST), with execution times often several or even 10 times longer than those of KUMAP. Although SITML takes the lead on certain specific datasets, its execution time surges abnormally on RetinaMNIST, indicating poor robustness. Meanwhile, KUMAP and its variant SKUMAP maintain relatively stable execution times across all datasets while

delivering high performance, which validates their optimised numerical stability.

4.6.2 | Memory Control Analysis

In terms of memory control, KUMAP and SKUMAP exhibit an overwhelming advantage on small-scale datasets. On four datasets: BloodMNIST, DermaMNIST, PneumoniaMNIST and RetinaMNIST, KUMAP and its variants secure the top two positions, with memory requirements substantially lower than those of UMAP and UMATO. For instance, on DermaMNIST, the memory footprint of KUMAP is only 61.72 MB, which is less than 5% of UMAP's consumption (1224.25 MB). Although the memory overhead increases on ultra-large-scale datasets like OrganAMNIST due to the inherent characteristics of kernel matrix computation, leading to it being surpassed by PaCMAP, it remains at a moderately superior level overall.

In summary, KUMAP and SKUMAP successfully strike a balance between computational speed and memory efficiency. By alleviating the heavy burden typically associated with high-dimensional mapping in traditional manifold learning, this series of algorithms remains highly effective for processing complex medical imagery in conventional computing environments lacking high-end GPU support.

5 | Conclusion

In this paper, we have proposed a novel kernelisation framework based on manifold learning to address the critical limitations of UMAP. Specifically, we introduced Kernel UMAP (KUMAP) and its supervised extension, Supervised KUMAP (SKUMAP). These methods successfully resolve the Out-of-Sample (OOS) embedding problem, which has long hindered UMAP's ability to project unseen points into a pre-learned embedding space without recomputing the entire manifold.

Our experimental analysis across eight diverse medical image datasets yields several key findings:

- *Out-of-Sample Robustness*: Both KUMAP and SKUMAP exhibit superior performance in embedding new samples, maintaining the global and local structure of the manifold more effectively than traditional manifold learning methods.
- *Discriminative Ability*: By incorporating label information, SKUMAP significantly enhances the separation of classes in low-dimensional space, leading to a marked improvement in classification performance for complex medical imagery.
- *Generalisation*: The kernel-based approach provides a mathematically principled way to generalise UMAP, making it a robust tool for real-time diagnostic applications where new patient data must be processed rapidly.

In conclusion, the proposed framework not only bridges the gap between manifold learning and kernel methods but also provides a practical and scalable solution for high-dimensional

medical image analysis. Future work will explore the optimisation of kernel selection to further reduce computational complexity in large-scale clinical settings.

Author Contributions

Benchao Li: software, writing – original draft, writing – review and editing. **Yun Zou:** visualisation. **Ruisheng Ran:** methodology, conceptualisation, supervision.

Funding

This work was supported by Chongqing Municipal Education Commission (KJZD-K202100505), Chongqing Municipal Science and Technology Bureau (cstc2020jcsx-msxmX0190), Ministry of Education of China (20YJAZH084).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Endnotes

¹ <https://medmnist.com/>.

² <https://github.com/MedMNIST/MedMNIST>.

³ https://github.com/scikit-learn/scikit-learn/blob/main/sklearn/discriminant_analysis.py.

⁴ <https://github.com/concavegit/kfda>.

⁵ https://github.com/scikit-learn-contrib/metric-learn/blob/master/metric_learn/nca.py.

⁶ https://github.com/scikit-learn-contrib/metric-learn/blob/master/metric_learn/itml.py.

⁷ <https://github.com/YingfanWang/PaCMAP>.

⁸ <https://github.com/eamid/trimap>.

⁹ <https://github.com/hyungkwonko/umato>.

¹⁰ <https://github.com/cvxgrp/pymde>.

¹¹ https://github.com/lmcinnes/umap/blob/master/umap/parametric_umap.py.

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